National University of Computer and Emerging Sciences, Lahore

IntelliLaw

Abdullah Nadeem 22L-6639 BS(CS)

Khawaja Muhammad Azan Saeed 22L-6749 BS(CS)

Muzammil Ahmed 22I-0970 BS(CS)

Supervisor: Ms. Aleena Ahmed

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Date: October 15th, 2025

Name: Abdullah Nadeem

Signature:

Name: Khawaja Muhammad Azan Saeed

Signature:

Name: Muzammil Ahmed

Signature:

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Abstract

IntelliLaw is an intelligent legal support and management application that utilises artificial intelligence to enhance access to justice and helps with the seamless handling of cases in Pakistan. It features a chatbot which uses the natural language processing to be trained on Pakistani legal materials, laws, and judicial cases that can give users the correct legal advice, related reference, and seek advice on the most common legal practices. IntelliLaw has also a full case management application to lawyers and firms which includes client information tracker, document storage, invoice generator, and hearing notes. Through finished language models and curated legal data, IntelliLaw will give the general population a first-line legal assistance and relieve legal practitioners of the administrative load, thereby enhancing efficiency, awareness and openness in the Pakistani legal system.

Executive Summary

The biggest issue in Pakistan is that the majority of the populace is not aware of their legal rights and, thus, they do not get much assistance of the system in terms of justice. Conversely, the majority of professionals in the legal field end up performing several dull monotonous tasks: the work on documents, the research of cases, and the clientele. This gap has been filled with the IntelliLaw that is an AI-based tool which has been created to help close this gap by ensuring that access to legal help by the people and lawyers itself has been made easier, more efficient, and reliable.

IntelliLaw is an AI legal chatbot-based platform, as well as a case management platform. Its underlying chatbot model is trained on Pakistani legal data, containing laws, judicial cases and protocols. This chatbot will respond to general questions of law and will attract attention to the corresponding sections of various acts such as the Pakistan Penal Code or the Civil Procedure Code, and be able to guide users through the process of filling a complaint one by one, defining and explaining their legal rights and even writing simple legal documents, such as contracts and affidavits. Thus, the chatbot is more of an initial step toward seeking assistance and provides users with information that they need to make decisions but is obviously not a replacement for professional legal advice.

The second part is the Case Management Module, which is designed for lawyers, attorney firms, and solo practitioners. It allows tracking of clients, document archiving, creation of invoices, sending of hearing reminders, and case progress tracking. This secure, web-based system enables professionals to manage multiple cases simultaneously and maintain their records in order. It improves productivity by managing and streamlining workflows, resulting in more satisfied clients due to reduction of paperwork and administrative expenses.

IntelliLaw was developed as a multi-phase, systematic process. First, datasets from trustworthy legal sources, such as pakistancode.gov.pk and pakistanlawsite.com, are selected, along with publicly available court archives. This information is then preprocessed to make the responses contextually appropriate and relevant under Pakistani law, and used to further improve existing NLP models. Openness is ensured through various disclaimers within the system to ensure that moral and legal requirements are fulfilled, such as the Pakistani Personal Data Protection Bill, regarding user privacy and the responsible use of AI.

Technically, IntelliLaw relies on advanced AI frameworks and Natural Language Processing techniques to ensure intelligent query understanding and context-aware responses. Its modular design ensures scalability, security, and easy adaptability for future extensions, such as increasing the number of legal domains to be covered or integrating the system with judicial databases. The main impact brought about

by IntelliLaw lies in its two benefits: it helps legal professionals by automating repetitive research and administrative tasks, and empowers the general public through free access to legal information. IntelliLaw bridges the gap between technology and law and helps Pakistan's legal system to be more open, effective, and accessible. In the long run, it can develop into a fully integrated digital platform of legal services aimed at creating greater awareness among the public and an increasingly technologically developed legal framework.

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Chapter 1 Introduction

In any given society, access to justice and legal awareness are two very important pillars of equality and good governance. However, in Pakistan, a large number of the population remains ignorant about their legal rights and the procedure to seek justice due to complex legal language, inaccessibility to authentic information, and unaffordable professional consultation. Simultaneously, lawyers and law firms experience a number of challenges with regard to workload management, organization of case records, and time-consuming research tasks. IntelliLaw is an AI-powered legal assistance and case management system that integrates Artificial Intelligence and Natural Language Processing to simplify access to legal information and streamline professional workflows. It therefore seeks to empower the citizens by providing effective legal advice and aiding legal practitioners with resources to manage their clients effectively, handle documents and make timetables. IntelliLaw also combats isolation and disconnect of technology and law so that both the interested parties can have a future where justice is more effective, clear and affordable.

1.1 Purpose of this Document

In this report, the design, development, and deployment of the Final Year Project, IntelliLaw will be described. The nature of our project is the creation of a legal assistant and management platform with an AI taking the center stage that will be used to create easier access to legal services and a day-to-day experience of the professional field within the Pakistani legal field. Its primary goal will be to build and create a viable and effective software application that will be able to combine the functionality of intelligent chatbots and an overall case manager.

It is an extensive specification that would possibly give the reader the motivation, goals, scope and shape behind the project. The methodologies, tools, and techniques that were utilized within the framework of the system development, including preparation of the dataset and integration of the NLP models, as well as interface design and creation of functional modules are to be detailed. The focus is on the practical construction of systems and not conducting theoretical research and experiments.

Through IntelliLaw, we would like to present a workable prototype which will demonstrate how Artificial Intelligence may be utilized to assist users to learn Pakistani laws, prepare documents depending on the legal mandatory, and handle professional case procedures in an efficient way. This report elaborates on the entire development of the IntelliLaw project, project vision, associated applications, software requirement, system architecture as well as high-level and low level design. This way, the document gives a lucid and well-organized description of the purpose of the project, the stages of its implementation,

and how it will contribute to making the sphere of law in Pakistan more accessible and efficient.

1.2 Intended Audience

This report will be aimed at the people and groups directly engaged in the analysis, interpretation, and application of the findings of the IntelliLaw project. It is primarily addressed to the evaluation panel, the project supervisor and the members of the faculty of the Department of Computer Science, National University of Computer and Emerging Sciences (FAST NUCES), Lahore Campus, to carry out the review of the system design, functionality and technical implementation.

In addition to the academic assessors, software developers, AI practitioners and researchers might be interested in the content of this report because they would like to have a further insight on how Artificial Intelligence and Natural Language Processing have been implemented in the field of law. The report records the methodologies and tools of development approach applied in the implementation of IntelliLaw and the major design choices that were made during the process. Thus, it has technical significance to new students or a team that works on similar projects.

Also, the report must be helpful to law professionals, law firms and other organizations that might be interested in using technology to increase efficiency on the research, documentation and handling of cases. An objectives, architecture, and implementation description outline of the project predispose makes the report an effortless reference to any person who desires to grasp how AI-based systems such as IntelliLaw can help in the modernization of the legal services in Pakistan and the-need to make more people access the services.

1.3 Definitions and Acronyms

1.3.1 Definitions

- **IntelliLaw:** An AI-powered legal assistance and management system developed in this project for the purpose of legal guidance and case management..

1.3.2 Acronyms

- **AI:** Artificial Intelligence
- **NLP:** Natural Language Processing
- **FYP:** Final Year Project
- **CMS:** Case Management System

- **API:** Application Programming Interface
- **GUI:** Graphical User Interface
- **MVP:** Minimum Viable Product
- **DBMS:** Database Management System
- **PDPP:** Personal Data Protection Bill of Pakistan
- **FAQ:** Frequently Asked Questions
- **ML:** Machine Learning
- **UI:** User Interface
- **UX:** User Experience
- **SDLC:** Software Development Life Cycle
- **SRS:** Software Requirements Specification
- **QA:** Quality Assurance
- **SDG:** Sustainable Development Goal
- **REST:** Representational State Transfer
- **ORM:** Object-Relational Mapping

1.4 Conclusion

This chapter outlined the overall concept and purpose of the IntelliLaw project, outlining the objectives, the intended audience, and some key terminologies that will be used throughout the report. In an effort to bridge the gap between technology and the legal world, the project seeks to offer a novel AI-empowered platform that will boost legal accessibility and professional effectiveness. The rest of the report is organized as follows: Chapter 2 Project Vision describes the motivation, goals, and scope of IntelliLaw; Chapter 3 Related Applications reviews existing systems and technologies relevant to the project; Chapter 4 Software Requirement Specifications outlines the functional and non-functional requirements of the system; and Chapter 5 High level and Low level Design presents the architectural and detailed design of the system along with the implementation approach. In Combination these chapters present the overall overview of the planning, development, and design process of the IntelliLaw.

Chapter 2 Project Vision

IntelliLaw has been developed to bring modernity and ease of the delivery of legal services in Pakistan by incorporating AI into the legal environment of this nation. The platform brids the disconnect between the legal professionals and the general population because it provides efficient, reliable, and accessible legal services via one digital platform. It incorporates a smart chat station that is capable of providing an answer to the most common inquiries in the legal field as well as case management platform that is used to structure the daily activity of lawyers and law firms. However, ensuring efficiency, transparency and access to the justice system, IntelliLaw offers access to the authentic legal information, case management and the possibility to create all the necessary documents on the same platform. This is one of the biggest steps towards digitizing the architecture of Pakistan and gives its citizens, as well as to the people running this business, power through technological innovation.

2.1 Problem Domain Overview

IntelliLaw is an AI-driven platform, which is aimed at facilitating legal information search and workflow-related management in the Pakistani legal field. The facility is offered by the system where the interpretation of the natural language is presented and the information concerning the Pakistani laws, judgment rulings, and documents are recognized. It assists the general population in knowing their rights and procedures involved by the law and aids the lawyers and firms with the tasks of client tracking, monitoring case proceedings, document storage and with the invoice generation, as well as hearing reminders. Intelligent query processing in line with effective management tools will ease the legal process, make it more accessible to everyone, and foster more efficiency in the legal environment of the country, IntelliLaw aims.

2.2 Problem Statement

The availability of legal knowledge and professional legal advice is one of the largest dilemmas that ordinary people have been experiencing in Pakistan. Most of them do not know about the laws that activate their daily activities, their legal rights and the process. Conceptualization of complex terms, the unavailability of genuine sources, and the inability to meet the high and unattainable prices of the services of professional help can commonly discourage individuals seeking justice. Lawyers and law firms experience difficulties with high workload, keeping customer records, and time-consuming administrative activities of preparing documents and memorizing hearing dates. The delays in court processes, decreased efficiency, and time wastage are caused by the inefficiencies.

The gap that exists is a digital solution that would assist the legal fraternity to manage their workload efficiently and to make the general populace interested in accessing correct legal information. IntelliLaw resolves this problem by providing a smart platform where case management features are united with the natural language interaction and legal information search in a single platform.

2.3 Problem Elaboration

Pakistan tends to leave the general population at a loss when dealing with its legal system that is too complex to be handled without professional assistance. Inefficient knowledge of rights and procedures to redress, through the complex formulations of the laws, limited centralized data as well as a failure to find a cheap source of assistance, keeps most citizens in the dark on their rights and legal processes of their redress. By doing this, one will end up being confused, misinformed, and wasting time in seeking justice.

Some of the issues that lawyers have in dealing with include caseload management, maintaining records of every client, drafted documents and manual scheduling of hearings. These are among the tedious administration processes, which decrease productivity and posing of human error.

The IntelliLaw seeks to address the concerns with an integrated digital platform comprising of two front liners, namely, refining the accessibility of dependable information on law by the populace, and simplify case management to lawyers. Its different sub-issues it serves include intelligent and intuitive delivery of legal information, support in querying, aiding in drafting of documents and client and case management systems. IntelliLaw is able to offer these requirements and therefore enhance the accessibility and efficiency of the Pakistani legal ecosystem.

2.4 Goals and Objectives

The aim of IntelliLaw is to offer solutions that can improve the access to legal information and also streamline the professional legal management in Pakistan. These goals form the basis on which this intelligent, user-friendly, and effective system for case management and legal aid shall be created. The following are the aims and objectives:

- To provide an intuitive platform that assists the public in swiftly and reliably receiving legal information and understanding their basic rights and processes.
- The objective is to design and implement an intelligent chatbot that understands the queries of the users and responds with accurate information based on Pakistani legal documents and laws.

- With the intention of helping users to create commonly used legal documents such as agreements, affidavits, and contracts.
- The objective is to develop an integrated case management system that will help attorneys and law firms store documents safely, manage clients, and keep track of case progress.
- To improve the efficiency of workflow through additional professional tools, such as client management, invoice generation, and hearing reminders.
- To ensure that the system follows all the privacy and ethical policies for maintaining data security in line with the Personal Data Protection Bill of Pakistan.
- To assist the legal fraternity in Pakistan in their journey of digital transformation to efficiency, ease of access, and transparency while narrowing the gap between technology and the law.

2.5 Project Scope

The scope of the project is to design a web-based system for case management and legal aid for both legal practitioners and the general public in Pakistan. This system will enable users to:

- Obtain specific answers to your legal questions, as stated in Pakistani laws and various legal documents, in simple English.
- Get information from reliable legal sources regarding rights, legal processes, and particular legal provisions.
- Prepare basic legal documents such as agreements, affidavits, and contracts using pre-made templates.
- Allow attorneys and law firms to securely store legal documents, manage clients, and track case status.
- Give the attorneys tools, including the client tracking, billing and hearing reminders.
- Offer a friendly web interface such that all the users will be able to use it freely and communicate with it conveniently.

Integration of the natural language processing technique into the contemporary web development technologies will be used in developing the project. The chatbot module will be trained on carefully selected datasets of Pakistani legal texts, rulings, and procedural laws for contextually correct and precise answers. The system will ensure a focus on privacy compliance and safety in data handling according to the Personal Data Protection Bill of Pakistan.

The list of deliverables will include the following:

- A helpful web-based case management and legal aid resource.
- A comprehensive user manual on the features and operation of the system.
- Documents regarding testing, implementation, and design of the system.

This project scope provides the roadmap necessary in guiding IntelliLaw towards attaining its objectives of bringing transparency, efficiency, and access to the legal system of Pakistan.

2.6 Sustainable Development Goal (SDG)

The IntelliLaw project contributes to the SDGs of the UN by fostering innovation, justice, and institutional effectiveness. In particular, the goals and features of the system address directly SDGs 9: Industry, Innovation, and Infrastructure and SDG 16: Peace, Justice, and Strong Institutions.

2.6.1 SDG 9: Industry, Innovation, and Infrastructure

SDG 9 focuses on building resilient infrastructure, promoting inclusive and sustainable industrialization, and encouraging innovation. IntelliLaw contributes to attaining this goal by providing a state-of-the-art technological solution to the legal industry of Pakistan. The system makes use of AI and NLP in the development of an online platform for legal aid and case management. IntelliLaw automates tasks, facilitates access to information, and hence supports the digital transformation of legal procedures, as well as the adoption of technology within professional practices for innovative and sustainable institutional development.

Figure 2.1.

Figure 2.1: SDG9 (Industry, Innovation, and Infrastructure.)

2.6.2 SDG 16: Peace, Justice, and Strong Institutions

SDG 16 aims to foster inclusive societies, provide access to justice, and strengthen institutional accountability. IntelliLaw contributes to achieving this goal by making the provision of legal information acces-

sible, improving public awareness about rights entitlements, and supporting citizens in understanding court procedures and processes. It presents efficient case management and documentation to the practitioners. By facilitating transparency, access, and efficiency, IntelliLaw improves the process of justice administration and the more broadly-related principle of equal access to justice.

Figure 2.2.

Figure 2.2: SDG16 (Peace, Justice, and Strong Institutions.)

2.7 Constraints

The implementation of AI concentration and NLP in the legal sphere has operational and technical problems. The quality and thoroughness of legal datasets are very necessary to provide the accuracy and success of a chatbot response. Prime importance and not easy is update and validation of legal information found in credible sources. Another weakness that should be resolved is the difference in user queries based on their language styles, complexity and legality as stipulated by law. The regulatory issues related to privacy of the data must be remembered and to be subject to the alignment with the Personal Data Protection Bill of Pakistan. This demands a close design and optimization to come up with a successful yet easy to navigate interface that satisfies the interests of the professionals and the general population.

2.8 Business Opportunity

The IntelliLaw has a huge business potential considering the growing preference of technology in both professional and legal circles within Pakistan. IntelliLaw will be able to position itself as a leading automation and support in law because individuals and legal firms are interested in getting more information, documentation, and customer relationship electronically. The system attracts a wide range of users such as independent attorneys, small firms, and the general public since it can provide immediate, reliable, and moderately low-cost legal advice. Moreover, it facilitates further value addition in subscription-based case management tools, integration with online law databases, and additional premium legal services due to its scalable structure. In essence, IntelliLaw offers excellent prospects for

long-term viability and commercial adoption as it will fill a real need in digital legal solutions.

2.9 Stakeholders Description/ User Characteristics

The general public: These are individuals seeking legal guidance, awareness, or clarification about their rights, laws, or basic judicial procedures. Most users in this group have little to no background in legal terminology, which often creates barriers to understanding their rights or navigating legal processes independently. IntelliLaw aims to bridge this knowledge gap by providing simplified explanations, step-by-step procedural guidance, and easy-to-generate legal documents. Users can draft simple applications, understand common legal terms, and prepare themselves before approaching a lawyer. This stakeholder group represents the largest segment of IntelliLaw's audience, as access to authentic legal information in Pakistan remains limited, outdated, or difficult for the public to interpret without professional assistance.

Legal Firms and Lawyers: Professional lawyers, individual practitioners, and law firms require tools that help streamline their daily operations. Their work revolves around managing clients, drafting documents, scheduling meetings, and organizing case files. Traditional manual approaches often lead to delays, mismanagement of data, or increased administrative burden. IntelliLaw supports this group by offering an integrated environment where legal professionals can automate repetitive tasks, track cases, communicate with clients, and access a centralized repository of relevant legal information. By reducing administrative load, lawyers can dedicate more time to complex legal matters, research, and courtroom preparation, ultimately enhancing their overall efficiency.

Administrators: System administrators are responsible for ensuring secure and stable system performance. They oversee user authentication, manage data accuracy, prevent unauthorized access, and ensure compliance with privacy regulations. Administrators also play a vital role in updating legal content, maintaining system uptime, and monitoring potential threats. Their responsibilities extend to ensuring that the platform remains reliable, up-to-date, and aligned with national legal standards. Without their continuous monitoring and improvements, IntelliLaw would not be able to maintain the integrity and trust required for a legal-assistance platform.

2.9.1 Stakeholders Summary

IntelliLaw will support the various interests of the entire legal ecosystem in Pakistan. The product is a productivity tool to practitioners as well as knowledge that people can readily access. The system meets the needs of each of the less-experienced end-users that need guidance and more-established practitioners that need workflow efficiency through incorporating the legal assistance with the management features. By doing so, it offers a single interface that can be useful in meeting the special demand of the

entire user groups.

2.9.2 Key High-Level Goals and Problems of Stakeholders

All the stakeholders major goals are based on accessibility, efficiency, and reliability. In case of the general population, the primary problem is that no legal information is available and understandable. IntelliLaw breaks down the content of laws, simplifies the connotation of the response and therefore gives quick and precise answers to questions. Lawyers and firms are primarily concerned with the issue of workload management, as well as client and case records. The case management module of the system will equip them with systems to automate, organize and streamline the same. The three objectives of administrators include the platform stability, data security and adherence with privacy laws. Together, the IntelliLaw will address any gaps in the legal sector by providing an easy to use, transparent, and technologically enhanced solution to all stakeholders.

2.10 Conclusion

The chapter gave the general overview of the IntelliLaw project and explained its problem area, goals, and objectives, and scope. It indicated essential stakeholders, associated Sustainable Development Goals, and significant system limitations and opportunities. All these factors helped to explain the purpose of the project and prepare the functional and technical specifications of the following chapter.

Chapter 3 Literature Review / Related Work

This chapter carries out a thorough discussion of the legal systems and technologies that have helped in shaping the IntelliLaw. It discusses different platforms and applications that can be used to utilize artificial intelligence, natural language processing, and digital case management to enhance efficiency in the legal industry. In this analysis the chapter indicates the essence, benefits and negativities of such systems and pinpoints the existing gaps that still persist in offering legal services that are accessible and locally applicable to Pakistan. The development of IntelliLaw is based on this review, where IntelliLaw is shown to build upon the already existing innovations to provide a rollable, hyper-localized, and usable solution to the legal practitioner as well as the population in general.

3.1 Definitions, Acronyms, and Abbreviations

3.1.1 Definitions

- **ROSS Intelligence:** The legal research assistant, an AI using NLP to process queries and find the corresponding laws and cases.
- **DoNotPay:** A chatbot which provides simple issues with automated legal advice.
- **Clio:** This is a cloud-based legal firm management application designed to track cases, do billing, and documents.
- **LawDroid:** This is a virtual assistant that will automate the communication with clients, their schedule, and intake of the case.
- **LISA (Legal Intelligence Support Assistant):** This type of tool is a smart document drafting system, which assists users to create contracts and agreements.
- **CaseFox:** An online time tracker, billing and client management software.
- **Legal Robot:** A computer program that helps to understand legal texts and make them compliant.
- **E-Wakeel:** A Pakistani site that links the user with lawyers to access the legal system even easier.
- **Lawrina:** Legal education website where a person can find templates and form generator in addition to legal education content.
- **GRAPE QA System:**A question-answering system that is augmented with the knowledge graph to make the system more accurate.
- **LegalZoom:** A digital company that provides generated documents and business strategy.

- **Rocket Lawyer:** This is a subscription-based legal document creation and attorney-consulting service.
- **Lex Machina:** It is a data analytics system that offers a view of the litigation tendency based on AI.
- **Casetext:** This is an artificial intelligence based legal search engine which is built on semantic technology.
- **CourtListener:** Law After Database This offers free court opinions and legal information.

3.1.2 Acronyms

- **AI:** Artificial Intelligence
- **NLP:** Natural Language Processing
- **SDG:** Sustainable Development Goal
- **UI:** User Interface
- **UX:** User Experience
- **API**: Application Programming Interface

3.1.3 Abbreviations

- **FYP:** Final Year Project
- **CRUD:** Create, Read, Update, Delete

3.2 Detailed Literature Review

3.2.1 ROSS Intelligence

With time, the areas of professional practice of any kind are slowly beginning to be redefined by artificial intelligence, and the legal one is not an exception. ROSS intelligence became one of the first tools that could create the actual inroads into the field as the system focused on simplifying and ensuring that legal research was much faster. Constructed to assist lawyers in navigating the complex case law, it has substituted the manual search with an intelligent engine that comprehends the plain English question. This opened a new age in the historical context of data accessibility to attorneys: as opposed to collecting information, they could play an active role in interpreting it.

3.2.1.1 Summary of the Development Project

ROSS intelligence [1] was also created as an intelligent digital research assistant that managed to comprehend the natural language requests and provide the correct legal answers within seconds. Rather than typing complex Boolean search queries, a user would just speak, and say, What are the legal precedents of breach of contract under New York law? and the system would search its data base to give the most pertinent decisions, laws, and case abstracts.

The platform was operated by the Watson technology created by IBM, which is a machine learning plus language understanding tool that identifies the context and intent. Important passages of the legal opinions were underlined by ROSS with related cases proposed subsequently, which greatly shortened the time spent on research. Law firms seeking faster and more precise research tools were the major users of it because it was part of their strategy to better cope with the flood of information as well as assist in promoting data-based approaches to legal decisions.

3.2.1.2 Critical Analysis (Strengths and Weaknesses)

ROSS intelligence was known to have great respectability in the application of pragmatic artificial intelligence in the field of law. Its plain language queries made it so easy to use compared to the conventional database systems and its speed made research very easy. Its abilities of handling large aggregations of cases were especially beneficial to mega-firms. Nonetheless, there were several shortcomings of the platform. It used licensed sources of data which meant that it was prohibitively expensive to small companies and individual lawyers. Also, its algorithms were very talented, but there were times that it did not understand the rich contextual nuance or jurisdiction-specific interpretation. These issues were compounded by a problem with licensing and eventually resulted in the cancellation of the project in 2021.

3.2.1.3 Relationship to the Proposed Work

Similar to ROSS Intelligence, IntelliLaw aims at assisting and facilitating the knowledge of law that employs intelligent systems. The goals of the two are also similar in that they aim at facilitating research and making the acquisition of legal information easier. The distinction here lies in the fact that IntelliLaw is more locally orientated, with functionality added to it. Specifically designed to fit the Pakistani legal environment, it incorporates an interface to communicate with an end-user by the integration of a chatbot. Whereas ROSS was more focused on law firms, IntelliLaw is focused on both professionals and regular individuals and integrates legal support with a case management application store. It is this broader conception that has helped IntelliLaw to offer more localized, extended, and viable solution to

the legal community of the country.

3.2.2 DoNotPay

Artificial intelligence is not arrived at by halting the transformation of professional legal practice, but instead it has created the possibility of the likes of the general population to seek legal assistance on their own. In this respect, DoNotPay is one of the most successful examples of this transformation. Referred to as "the world's first robot lawyer," this platform made it possible for ordinary users to resolve everyday legal issues without relying on costly legal representation.

3.2.2.1 Summary of the Development Project

Founded in 2015 by Joshua Browder, DoNotPay [2] was originally a free online chatbot that assisted users in disputing parking tickets in the United Kingdom. From humble beginnings, it grew to cover numerous topics such as consumer protection, refund processes, immigration aid, and minor court filings. The system works in a direct manner: one types in the problem as simply as possible, and then the chatbot prepares the documents or even step-by-step instructions for fixing the problem.

The biggest mission undertaken by DoNotPay is making access to legal aid cheap and available for anyone. Its intuitive interface, automated templates, and conversational flow allow any user to independently navigate minor legal issues. The site eventually achieved massive momentum and retains millions of users in various countries, and the way automation can bridge the gap between the law and people.

3.2.2.2 Critical Analysis (Strengths and Weaknesses)

DoNotPay is beautiful because of its simplicity. It removes the fear component that comes along with legal procedures, and all users have is a natural means of sharing their issues and receive real to life assistance immediately. This is very handy in the case of a common user since the system is scalable and less expensive. The limitations of it, however, are not very difficult to grasp. It is functioning under a template, so it is clearly not able to handle a case of complex and nuanced needs requiring professional interpretation or subtle resolutions. There is also a problem of differences in jurisdictions. These inadequacies are also a clear desire towards a more flexible, situation-sensitive system that is not overly automated but employs accuracy at a higher local rate of response.

3.2.2.3 Relationship to the Proposed Work

DoNotPay and IntelliLaw have the same vision of providing access to law to everyone. But IntelliLaw predicts that they can implement this dream on the Pakistani legal landscape. IntelliLaw will provide intelligent query support with full-fleet case management to lawyers, rather than just document generation.

This bipartisan approach allows it to benefit both the citizen and the profession on a unified platform a balanced ecosystem that can effectively assist in empowering the populace and ensuring efficiency in the law at once.

3.2.3 Clio

The legal industry has been rapidly adopting the digital revolution, but in that regard, the technology has remained the firm partner in the control of the daily activities at the law firms. Among the more established tools available is Clio which is a platform that has been mentioned to transform the way practitioners arrange, chart and conduct business. Clio has pioneered the modern-day management of the legal practice across the globe by integrating case handling services, client communications as well as billing into a single online platform.

3.2.3.1 Summary of the Development Project

Clio [3] It is an online cloud solution of managing the entire workflow of a law firm, including time tracking, document management, billing, and client communications and case management. It assists the single user or hundreds of user firms comprising of multiple clients and multiple cases by providing convenience in appointment, unsecured access of files and storing information in one location.

Having been established in 2008, Clio has emerged as one of the most successful SaaS systems in the legal business. Its cloud based system enables attorneys to work anywhere but at the same time work in a team. Convenience is also increased with the help of integration with other tools, including the Microsoft Office and Google Workspace, which makes Clio a convenient and valuable tool used by law practices.

3.2.3.2 Critical Analysis (Strengths and Weaknesses)

Clio is well praised by its feature set and functionality. It makes it easier to run a law firm in that all the operations are from one system which can be easily accessed anywhere. This in turn increases productivity to a high degree because of improved accessibility and arrangement. There is a negative side to this, where Clio is more aimed at professional usage and is not likely to fit individual practitioners or small firms due to the subscription. It might also be limiting because it requires stable internet access which might not be easily accessible in places where such is limit. In spite of these restrictions Clio continues to be one of the most useful and stable legal management programs in the market.

3.2.3.3 Relationship to the Proposed Work

Although Clio is chiefly concerned with law firm management, IntelliLaw is a firm that performs some fundamental organizational concepts but applies them to the legal atmosphere of Pakistan. The similarity is that both of them are concerned in the effectiveness of the workflow and the structured recordkeeping, yet, IntelliLaw goes a step further, integrating a conversational assistant that can assist the professionals as well as the general audience. It thus transforms IntelliLaw into a productivity tool and a legal support system, building on top of the functionality such as Clio but in a much broader and more community-driven paradigm to fit needs on the local level.

3.2.4 LawDroid

In fact, automation is now a thing in the practice of law and LawDroid has played a vast role in bringing tremendous change in that subject. LawDroid is a legal assistant that comes in the form of a conversation, with its focus being on automating repetitive administrative duties and enhancing communication between law firms and the clients. It is an indication of increased trend to utilize digital assistants to save time and improve the service of clients in the legal practice.

3.2.4.1 Summary of the Development Project

LawDroid [4] is an online lawyer assistant system through which lawyers can interact with clients, bookings and even documents through automated systems. With the help of custom chatbots, companies could create thread of automated dialogues, which could solicit client data, deliver updates, or even create rudimentary documents automatically. This would facilitate in minimizing manual activities and enable the legal professionals to concentrate on high level activities. It also will readily integrate with other case management tools available to the market to ensure uniformity in data as well as come up with an efficient system of communication between the two cases. Having a capability to manage several client dialogues simultaneously renders it particularly helpful in the busy companies that prefer digitalizing communication without the need to lose personalized service.

3.2.4.2 Critical Analysis (Strengths and Weaknesses)

The key benefit of LawDroid is that it can not only automate the process with external customers but also, consequently, leave lawyers without the monotonous routine that concerns the administrative side of work. The fact that it can be integrated and the design of the chatbot can be customized makes it flexible to various practices. Nevertheless, in essence, the system is based on pre-determined answers, and it is not entirely responsive to the complex legal logic and case-specific examination. The set up

process can also be complicated by smaller firms having low levels of technical expertise. Nevertheless, despite this platform, it is possible to fifth the benefits of automation in increasing communication, productivity, and efficiency in managing a legal operation.

3.2.4.3 Relationship to the Proposed Work

The implicit similarity that exists between the application of conversational technology to enhance legal processes between IntelliLaw and LawDroid is that they both are interested in efficiency by automation of the work. The IntelliLaw just goes a few steps further and applies the legal support to the civilians in Pakistan as well. In addition to serving professionals, it enables citizens to seek clarifications regarding national laws and get advice on the same. The case management module that is built in it also provides a more integrated approach that links both clients and lawyers to one digital platform.

3.2.5 LISA (Legal Intelligence Support Assistant)

With increased needs to access inexpensive services with regard to creation of legal documents, digital platforms have begun to automate the procedure of creating legal documents. LISA is one of the oldest and most successful systems in this regard and makes a good example of how smart automation can help the user when it comes to creating legally sound agreements without requiring the input of a lawyer, directly.

3.2.5.1 Summary of the Development Project

LISA [5] is an automation document platform that assists in the preparation of legal documents during confidences agreements, employment and partnership deals. The system utilizes interactivity technique through questionnaire; it takes in the input of the user and automatically generates a customized document, depending on the response of the user. This process is guided and assists the users to come up with correct and comprehensive legal forms with less chances of missing crucial clauses.

LISA is a software which is rule-based with natural language understanding skills to enable a non-expert user to prepare professional legal documents in an efficient manner. Its primary goal is to ensure that going through the legal drafting process becomes more convenient, cheaper, and not as fearful to those who are not trained professionals in the area of law.

3.2.5.2 Critical Analysis (Strengths and Weaknesses)

LISA has the largest strength in being easy to use and reliable. It enables the people having no legal background at all to develop formal documents that comply with the basic standard requirements and, therefore, save time and consultation rates. The question and answer method is guided, so it is not that

vague and thus likely to cause human error is minimal. It is however limited in its flexibility because its dependence on fixed templates does not allow flexibility when it comes to those contracts that are specialized or region-specific. It is not the answer to professional juridical recommendations on more complicated cases, but LISA was a significant move in available, affordable legal automation.

3.2.5.3 Relationship to the Proposed Work

The overall vision remains the same: to improve public access to legal services through technology. The only difference is one of scope and localization. IntelliLaw expands on LISA's document automation concept by bringing in legal information retrieval and professional case management within the Pakistani legal system. It also adapts templates and responses with respect to local laws, hence making them more functional for a regional user. In doing so, IntelliLaw is holistic in that it brings together both the concept of public assistance and the tools legal professionals use on one platform.

3.2.6 CaseFox

Law firms nowadays often have to deal with multiple clients, several cases, and simultaneous billing activities. All these tasks are considerably time-consuming and may invite errors if done manually. CaseFox is an application that was developed as a practical all-in-one solution to make legal professionals' administrative work easy. Since then, it has been one of the preferred tools among lawyers in their day-to-day activities to obtain efficiency, accuracy, and transparency in their daily operations.

3.2.6.1 Summary of the Development Project

CaseFox [6] is a web-based management tool designed to help law firms and solo practitioners organize workflow. Users can record billable hours, track case progress, generate invoices, and store client data in a totally secure web environment. All case-related documents, notes, and communication can be accessed from one dashboard, thereby making coordination more smooth and reducing the load of administrative work.

The platform also features expense tracking, document sharing, and customizable reporting, all of which enhance accountability and financial management. CaseFox has been made easy to use, even to those with no prior technical experience. Over time, it has proved to be a reliable system by the companies of any size to have a better relationship with their clients in terms of billing on time and established records.

3.2.6.2 Critical Analysis (Strengths and Weaknesses)

CaseFox is regarded as efficient in streamlining the management of the legal routine. It is user-friendly, includes billing modules, and allows easy customization of reports, becoming relevant to companies that have to closely measure productivity and costs. It is cheaper and more affordable compared to other enterprise-level tools. Nonetheless, it has too large dependency on internet connectivity, and its visual effect resembles outdated versions more than the latest options. Nevertheless, it is a trusted and useful choice to the legal professionals seeking a consistent case management tool.

3.2.6.3 Relationship to the Proposed Work

IntelliLaw, like CaseFox, is aimed at increasing organization and productivity amongst practitioners. The two platforms reduce manual work and facilitate smooth processing of the work. Nevertheless, IntelliLaw is not limited to administrative matters, but it also presents an intelligent legal assistant that provides the users with the appropriate legal information and support in matters related to the legal system in Pakistan. Also, it combines artificial intelligence and localized functionality and thus allows IntelliLaw to serve a dual purpose as a productivity platform and as a legal support system, which is publicly available and operates to remedify the shortcomings of other traditionally used management tools.

3.2.7 Legal Robot

Most of the records in the legal field are in a highly complex form, which is difficult to understand by individuals who do not have legal background. In order to make legalese less ambiguous and understandable, Legal Robot was created, which is a web application that rates contracts and other legal texts using plain English. It is indicative of a critical trend towards openness and availability of legal documents.

3.2.7.1 Summary of the Development Project

The website Legal Robot [7] is a web application, which examines uploaded legal texts, their legibility, fairness, and compliance with established standards of legal writing. It detects missing clauses, ambiguity, or other inconsistencies which can present a risk to a user. It opposes uploaded documents with the rest of examples in its database that allows users to understand how their contracts are different than standard legal practices.

The tool applies natural language processing and machine learning, which are trained on very large amounts of legal text data. The tool allows simple, non-complicated communication of results in share-

able language, therefore, making the individuals and businesses cognizant of what they are committing to without the need to acclimate themselves to the professional legal advice in entirety. The main objective of Legal Robot is to provide individuals with integrity and understandability of working with legal documents.

3.2.7.2 Critical Analysis (Strengths and Weaknesses)

The originality of the Legal Robot lies in the fact that it simplifies and makes the legal text easier to understand. It reduces confusion by simplifying technical jargon by plain terms and also helps the user to establish the areas which could pose some difficulties. Its benchmarking capability also advances this by the indication of the extent of a contract portraying accepted norms. However, its precision could vary by jurisdiction, because it is actually conditioned by the data of the English language law. It also is not able to fully substitute the professional review in complicated or high risk issues. It is inside these boundaries that Legal Robot makes a significant contribution to fairness and comprehension in legal communication.

3.2.7.3 Relationship to the Proposed Work

The interest of Legal Robot in streamlining legal data is related to the overall goal of the IntelliLaw making the law accessible to everyone. The two systems minimize reliance on the existing law intermediaries, whereas IntelliLaw goes a notch higher to contextualize its advice with Pakistani legislation and provides real-time communication in its chatbot. Although Legal Robot specializes in document analysis, IntelliLaw offers an integrated platform, which is oriented to understanding, usability, and management in a local environment.

3.2.8 E-Wakeel (Pakistan)

In Pakistan, digital transformation has gradually penetrated into the legal sector with tools that enhance the interactions of the citizens with legal professionals. Among the first and the most prominent ones, it is recommended to mention E-Wakeel- a site which was created to enable people to contact a lawyer and arrange a communication without any problems online. It is also a symbol of the trend of the nation to become more technological in terms of legal assistance.

3.2.8.1 Summary of the Development Project

E-Wakeel [8] is an online platform which fills the gap between the customers and the lawyers. The users have options to search professionals by their area of expertise, city or affiliation with a court. There is also an opportunity to book a consultation online, and informative articles and frequently asked

questions are also available on the platform so that the user could get a clearer idea of the legal processes before calling a lawyer.

E-Wakeel was developed with the emphasis on convenience and trust as the people are able not to use personal references to reach registered and trusted lawyers. It transforms legal interaction (especially in direct digital exposure) into a more transparent and structured interaction, particularly to users who were previously not able to receive reliable legal means of assistance.

3.2.8.2 Critical Analysis (Strengths and Weaknesses)

The most significant strength of E-Wakeel is that it is a localizedpectacle. Particularly, it will cater to the credibility of Pakistani case law by providing an authentic registry which will directly couple the clientele with attorneys. The fact that it provides online consultation and verification system builds more trust and credibility. On the other hand, the platform primarily functions as a listing and communication platform and lacks interactive legal advice and automatization. Its effectiveness over a long period is conditional on periodical updates and involvement of the user. Irrespective of these shortcomings, E-Wakeel is a crucial step in making change and transforming the legal system in Pakistan by digitalizing it.

3.2.8.3 Relationship to the Proposed Work

E-Wakeel provides the basic model to IntelliLaw, they both aim at providing more people with access to legal services. Although E-Wakeel aims to find a match between a client and a lawyer, IntelliLaw takes the concept a step further which involves the incorporation of artificial intelligence and natural language processing to provide real-time legal services and document automation. It is also adding a case management system to legal practitioners, comprising of a single, and smart platform, which is of advantage both to the client and the practitioner. This way, IntelliLaw transforms the concept of E-Wakeel into a more progressive and all-encompassing digital in the legal ecosystem.

3.2.9 Lawrina

With the shift to online legal services, other platforms like Lawrina have become the simple and reliable resource that an individual and a small business can use to access affordable and appropriate legal services. Lawrina is a convenient service through which a non-lawyer can conveniently generate the necessary legal papers and understand its contents without having to engage the entire professional support. It simplified daily legal tasks of users who only need accurate well structured documentation by being able to mix both convenience and technology.

3.2.9.1 Summary of the Development Project

Lawrina [9] will be an online service where its users create and tailor legal papers within guided templates. Its system supports a wide variable of document type, such as lease agreement, the employment and non-disclosure agreements. The user responds to a limited number of form-based questions, and the automatic generation of documents based on the specifics of the information is received.

Additional to creating documents, Lawrina provides a wide range of online legal articles, clarifying notes, and how-to manuals which can be used to comprehend the meaning and context of their documents. This automation with educating process enables users to perform general legal operations effectively and at a very small fraction of the time and money spent in the traditional legal services.

3.2.9.2 Critical Analysis (Strengths and Weaknesses)

The most important strength of Lawrina is the simplicity and affordability. The templates have been reviewed by professionals and are therefore good in common legal situations but on the other hand it is user friendly therefore anyone can use it even without having prior legal background. This does not, however, suit the flexibility of country-specific laws, and it does not have higher-end features like real-time consultations or case management. Nevertheless, Lawrina can be considered a viable option to use in common legal procedures and legal literacy as it is still a reliable gateway to virtual legal services.

3.2.9.3 Relationship to the Proposed Work

The mission statement of the philosophy behind Lawrina conforms well with IntelliLaw mission to make law more accessible and comprehensible. The two promote access to legal information through self service, IntelliLaw expands on this with the inclusion of an intelligent chatbot and built in case management system specifically designed to meet the needs of the Pakistani legal system. Instead of providing document generation equipment with a pre-defined set of templates, IntelliLaw combines document creation and interactive-context-related guidance. Such a dynamic and localized model contributes to the fact that IntelliLaw is a more inclusive and holistic platform of handling daily legal requirements.

3.2.10 GRAPE QA System

The development of the artificial intelligence has introduced another generation of the question answering systems that contribute to contextual understanding and are able to deliver factually correct answers. In this direction is the creation of the GRAPE QA System, which involves the graphical implementation of knowledge into natural language processing in order to provide improved machine interpretation and responses to open-ended questions. This project demonstrates that the structured knowledge improves

the thinking process and preconditions the use of special functions in particular fields such as the legal one.

3.2.10.1 Summary of the Development Project

GRAPE (Graph based Passage Reader for Open domain Question Answering) [10] model is aimed at relating the unstructured text to structured information to generate right and contextually appropriate responses. It extracts collegial passages of any text collection of large size and constructs a semantic net that describes how entities referred to in any text relate to each other. This enables it to get the more insightful clue of an inquiry rather than just matched the key words.

GRAPE can be considered as a meaningful change in the search behavior as the superficial search is replaced by the actual understanding of the concept, which is a synthesis of machine learning and semantic thinking. Since it allows linking the fragments of information provided in various documents, it assists in creation of answers that are in turn both meaningful and verifiable; therefore this is a significant move towards the smart developments of information systems.

3.2.10.2 Critical Analysis (Strengths and Weaknesses)

GRAPE can be extremely useful in relation to improving the reliability of automated question answering by incorporating both structured and unstructured knowledge sources. This is possible due to its modular structure, which allows it to change and be adapted in the broadest possible range of fields, including science and education, law and policy. Nonetheless, it needs high quality and domain specific training data to work well; this restricts its use in those regions where such data is not available. System deployment is also expensive to the smaller organizations due to computational requirements. However, despite these issues, GRAPE can be taken as one of the examples of how a mix of deep learning and the semantic net can lead to numerous contributions to machine cognition.

3.2.10.3 Relationship to the Proposed Work

The underlying concept of GRAPE parallels The inherent idea behind GRAPE is similar to IntelliLaw in that it reaches an accurate response based on the context of an intelligent data processing. Though GRAPE is a broad open domain structure, IntelliLaw has applied the same methods to a well-delimited legal domain. Training IntelliLaw applies Pakistani laws, rules and case law data to develop its system, modifying the strengths of GRAPE in contextual reasoning and complex structure knowledge synthesis to a specialized and real-life field. The outcome is that it is possible to respond to the legal queries of the users reliably but with local relevance.

3.2.11 LegalZoom

LegalZoom has become one of the most famous and popular online law websites with consumers and small businesses across the past decade. It was also to bring legal assistance closer to the people, quicker and less expensive, particularly those individuals who may not necessarily have easy access to a law firm. Integrating both automation and templates reviewed by experts, LegalZoom has led the frontline of defining the way the user community, in general, prepares documents and consults with lawyers on the internet.

3.2.11.1 Summary of the Development Project

LegalZoom [11] is an online web portal providing information-sharing platform in which users are guided through the process of creating legal documents such as business incorporation documents, wills, intellectual property filings, and even contracts. It is a step-by- step guided process with questionnaires that archive the user information and automatically generate user-specific documents.

Other than automation, LegalZoom also offers its users licensed attorneys to consult and receive subsequent help in compliance business issues. The secret of its achievements lies in its design oriented on the reliability and convenience: each template of the document has been tested by legal experts beforehand and its interface helps the user follow each subsequent step. It has become the reliable option among millions of people in need of basic and cheap legal support.

3.2.11.2 Critical Analysis (Strengths and Weaknesses)

Yet the area where LegalZoom really sells is its coverage and size of library. It makes readily available at a relatively low cost legal advice to a segment of the market that would not otherwise solicit it. The majority of its content is of U.S. law, however, making it less valuable to the rest of the world. Second, its templates are good, albeit there is no adaptive reasoning-the system merely dies off with a system output that does not necessarily reflect the user intents. Nevertheless, LegalZoom is one of the brightest examples of how technology can help scale legal access to the audience in spite of the mentioned limitations.

3.2.11.3 Relationship to the Proposed Work

LegalZoom and their vision of simplifying one of the most difficult areas in legal practice is closely echoing the goals of IntelliLaw because they both simplify the process of accessing legal services and make them cheaper. It is only that IntelliLaw is designed to manage the legal context of Pakistan, integrating natural language processing (hypothesis conversational support). IntelliLaw combines document

generation automation with smart legal advice incorporated into a cases handling system, and follows the trailblazed approach of LegalZoom in developing a legal ecosystem based on its use-friendly philosophy. In such a manner, it will help close the divide between the general and the professional legal community in a manner that is locally appropriate and technologically high.

3.2.12 Rocket Lawyer

Rocket Lawyer belongs to the group of the oldest online services that exist to make simple legal tasks easy, whether to an individual or a small business. The site mainly offers users the functions to accomplish routine legal activities like writing a contract or consultancy services without entirely depending on counselors. Over the years, Rocket Lawyer has emerged as one of the most pertinent guides on how with the ongoing advances in technology, affordable, efficient, and widely available legal assistance goes. Presentation of the Development Project.

3.2.12.1 Summary of the Development Project

Rocket Lawyer [12] offers a unified online platform that enables preparation of documents, consultative legal services, and management of the legal needs. It provides customizable forms of agreements, leases, and incorporation documents of companies. The user is provided with an automatically generated, legally sound, document upon the answer to guided prompts. One of the peculiarities of the Rocket Lawyer is an opportunity to hire the lawyer on demand, providing the subscribers with a chance to use the services of the professional at a fraction of the price.

The subscription plan provides a renewed access to e-signing features, templates of the site, cloud storage, and reminders of the deadlines. A combination of these functions forms an entire ecosystem of handling your legal affairs online, a model which has played a significant role in the current digital revolution of the legal industry.

3.2.12.2 Strengths and Weaknesses (Critical Analysis)

The principal strength of Rocket Lawyer is the combination of automation with professional assistance. On the one hand, users get highly competent legal counsel with little possibilities of paying extremely high fees at the same time its clear interface is attractive to individuals who have no legal background. Nevertheless, the scope of this platform is quite limited to the jurisdictions of the U.S. and U.K., which poses restriction on the utility that this platform may have in other locations. Its subscription charges can also deter users in the low income areas and its responses to its users is also based on fixed templates since it does not have much contextual grasp. However, it is true that Rocket Lawyer is an effective model of incorporating human service to digital efficacy in offering online legal services to the marital

proceedings.

3.2.12.3 Relationship to the Proposed Work

The moderate automation and expert services offered by Rocket Lawyer resonate with the goals of IntelliLaw. They both strive to make legal assistance available and reliable, but IntelliLaw uses the concept in regard to the Pakistani legal environment. IntelliLaw provides case specific advice using an AI driven chatbot and built-in case management software, which adapts the advice to the jurisdiction. In addition, they have made it an inclusive site that can accommodate both professionals and the common man in the society as opposed to Rocket Lawyer and ensure that technology can help in enhancing fair access to justice in Pakistan.

3.2.13 Lex Machina

Lex Machina will be one of the first to include the use of AI and data analytics in legal practice. It has been designed to provide lawyers and companies with information-based guidance on proper strategy-making. It gathered litigation information and converted it to actionable intelligence which would convert legal research to a manual research procedure to an informed predictive undertaking.

3.2.13.1 Summary of the Development Project

Lex Machina [13] is a legal analytics sector that gathers court data in an organized way and evaluates it to provide it in a usable format. It provides an analysis of the tendencies that describe the judicial decisions and estimates the case outcomes and the law firm performance. It identifies trends and risks through structured court records by applying a stronger natural language processing and machine learning algorithms to aid lawyers in predicting the data and analyzing risks.

Its interactive dashboard and visual reports also allow the user to examine the tendencies of the judges, the opponent behavior or what might occur in a case. Lex Machina would deliver strategic benefits of litigation to previously inaccessible legal data sets by converting mass volumes of legal information into easily digestible information.

3.2.13.2 Critical Analysis (Strengths and Weaknesses)

However, most important thing about Lex Machina is its predictive power. It allows the lawyers to make well-informed judgments as opposed to acting out of habit. Complex analytics is also made easy on the intuitive dashboards that enable the ease with which information can be interpreted. Nevertheless, it is primarily serving large companies and corporate departments and this makes it inaccessible to smaller practices and personalities. This immensity on US litigation data makes it less applicable in

other jurisdictions and countries with minimal digital case archives have a problem with integration. All these do not diminish the fact that Lex Machina is a strong case of how analytics can guide and enhance the legal strategy.

3.2.13.3 Relationship to the Proposed Work

Whereas Lex machine uses AI in the context of legal information of large scale, IntelliLaw aims to introduce similar technological smartness to the Pakistani legal system. IntelliLaw does not rely on predictive analytics only but also integrates information retrieval as well as chatbot support options and case management utilities which are beneficial to both citizens and practitioners. Thus, the IntelliLaw concentrates on repurposing the analytical center of Lex Machina but orienting it at access and inclusion on a local scale.

3.2.14 Casetext

Among the most powerful AI-powered legal research systems is the Casetext which is aimed at helping lawyers in their research even more speedy and more accurate. It transformed the process of locating and comprehending the relevance of case law by lawyers by developing highly complex search algorithms that have the capability to use natural language understanding. Casetext has established a niche as one of the most successful companies in the legal technology industry since its founding and rapidly adopted new models of AI and user focused processes that simplify the complex job of research. Its popularity has worked in wide welcome of AI in the traditional sluggish accepting occupational landscape of the legal community.

3.2.14.1 Summary of the Development Project

Casetext [14] is an intelligent search engine that expands what can be done regarding the practice of law research. The queries typed in plain English will give specific results that have been extracted out of cases, statutes and briefs; the semantic search can take it a step further than the previously used tools and read meaning rather than an equivalent of keywords matching. The system does not only pull documents but gets the contextual relevance of each source so that the user receives only the material that is of relevance to the legal argument he or she is pursuing.

The site contains an extremely large collection of American legal cases and allows users to add their briefs in order to find similar cases. Among many others, these aspects enable lawyers to present a better argument in a reduced amount of time. The creation of legal documents, summarizing and reviewing of them is also possible in Casetext; all in the same platform thus providing a much more integrated workflow to a practitioner. The first large scale legal research with semantic analysis was possible with

the Casetext, making it more efficient and context-sensitive than any previous research.

3.2.14.2 Critical Analysis (Strengths and Weaknesses)

Its significant strengths are that it is accurate, fast, and easy to use. The Casetext semantic search is better at increasing the quality of results and reducing the amount of research time. The user-friendly interface of the system reduces the learning curve of new users and the familiarity of seasoned practitioners with the system increases their productivity. The other weakness is that it depends on the availability and organization of U.S. legal data; thus, jurisdictions lacking an equivalent amount of digitized or inconsistently reported case law are unable to be supported equally. It might be limited in its geographical scope, yet Casetext remains on the heights of AI-based systems regarding legal research and proves the transformational nature of intelligent search tools.

3.2.14.3 Relationship to the Proposed Work

The method of semantic search of Casetext is vastly close to IntelliLaw in enhancing the legal research through smart automation. They do so in their own way, trying to make information about law more accessible and comprehensible. IntelliLaw takes the idea a notch higher since it is able to customize its database and responses by using Pakistani laws. IntelliLaw attempts to tackle the special issues of a legal system that is still evolving in the digital age, and whose specialists are more than frequently confronted by loopholes in available precedents. Its combination of chatbot support system, context-driven analysis, and case management makes what Casetext had done to U.S.-lawyers in its region applicable to Pakistan professionals and the population as well.

3.2.15 CourtListener

CourtListener is a legal database that is open access and is created to give free access to court opinions and other related documents. The Free Law Project has developed it with the intention of introducing openness and democracy to the legal system by removing obstacles associated with paid services to research. CourtListener has over time become one of the most preferred sources of information to legal researchers, lawyers and even citizens in the journey of seeking the knowledge of judicial ruling.

3.2.15.1 Summary of the Development Project

CourtListener [15] takes opinions of the federal and state courts and puts them in a single searchable database in the form of millions of decisions. With advanced filters and notification tools, the user is able to look into the text, citations, and metadata of a case. Citation tracking and sending of alerts to new cases added on certain topics are also available on the system.

However, in addition to legal research, CourtListener has also been used as the basis of open-data projects: its open APIs can be used by other developers and institutions to develop new applications based on the free court data. This has contributed to making CourtListener useful to both the legal community and academic and policy scientists.

3.2.15.2 Critical Analysis (Strengths and Weaknesses)

The most superlative feature that CourtListener has to offer is commitment to transparency and access by the people. It makes legalknowledge open and has no paywalls, fosters innovation in leg-tecthes and-because it is open API-based-is-allows collaboration and data-based projects. The contents though are generally restricted to the material of U.S. courts which makes it less valuable to foreign readers. Second, it does not have intelligent search capabilities that assist non-experts to locate materials of interest to them. Despite these points, CourtListener remains a needed example of transparency and visibility regarding legal research of the present-day.

3.2.15.3 Relationship to the Proposed Work

The IntelliLaw mission statement is highly aligned with the CourtListener principle of free access, which aims to give every person access to credible legal information. Whereas CourtListener has been offering unrestrained access to court information, IntelliLaw is offering interactive intelligence on top of access and provides a localized knowledge of the law. This synthesis makes the dead existence data to become active, an orientation to legal action.

3.3 Related Work Summary Table

Through the chapter of review on the available AI-driven legal platforms and technologies the world is making huge steps towards automation and digital transformation of the legal services. Nonetheless, every of the reviewed systems provides something different in the form of the most significant issues in the field of law, such as smart research and document automation, customer relationship management and access to justice by everyone. Most of the platforms covered are still limited based on geographical coverage, subscriptions or even just would not suit the Pakistani legal climate. This review preconditions the situation when it gives an idea about the latest technological environment and throws light on the gaps that the IntelliLaw platform is trying to bridge. A summarized table of the shapes analyzed below provides the main aspects of the systems, their applicability, and constraints in regards to the suggested solution.

The compact summary of Related Work is provided in the table 3.1.

Table 3.1: Related Work Summary Table

Application	Features	Relevance	Limitations
ROSS Intelligence [1]	AI supported legal research and case law discovery	Shows how intelligent search tools can improve efficiency in legal research and access to precedents	Built for Western jurisdictions; not adapted for Pakistani laws or general public use
DoNotPay [2]	Conversational chatbot for legal self-help	Demonstrates how automation can assist individuals with minor legal issues	Works only for predefined scenarios and lacks flexibility for regional or complex cases
Clio [3]	Digital case and client management platform	Highlights the role of cloud systems in streamlining law firm operations	Developed mainly for large firms; costly and not optimized for local legal processes
LawDroid [4]	Virtual assistant for client communication and scheduling	Shows how chat automation can reduce administrative tasks in law firms	Limited to predefined workflows; lacks localization for Pakistan's legal procedures
LISA (Legal Intelligence Support Assistant) [5]	Guided document creation and drafting	Demonstrates practical use of automation in contract generation	Provides only standard templates; no support for Pakistan specific laws or adaptive drafting
CaseFox [6]	Cloud platform for case, billing, and time tracking	Reflects how legal practices can manage documentation and billing online	Lacks intelligent automation and regional compliance integration
Legal Robot [7]	Automated contract review and readability check	Simplifies legal documents by evaluating clarity and fairness	Limited to text analysis; does not provide interactive assistance or legal advice
E-Wakeel (Pakistan) [8]	Web portal connecting clients and lawyers	Addresses accessibility issues within Pakistan's legal services	Works mainly as a lawyer directory; lacks automation or AI based legal interaction
Lawrina [9]	Legal document templates and educational guides	Offers easy to use resources for routine legal documentation	Not customized for regional laws; lacks smart legal interaction or guidance
GRAPE QA System [10]	Knowledge graph based question answering	Demonstrates how structured data enhances accuracy in information retrieval	Requires complex datasets and is not specialized for legal or local domains
LegalZoom [11]	Online document creation and business registration	Makes professional legal support accessible through digital forms	Restricted to U.S. laws and subscription users; not adaptable for developing regions
Rocket Lawyer [12]	Subscription based legal document and consultation service	Combines document automation with affordable lawyer access	Limited to certain jurisdictions; not tailored for Pakistan or public access
Lex Machina [13]	Litigation analytics and predictive case insights	Illustrates how legal data can support strategic decision making	Serves mainly corporate firms; unsuitable for smaller practices or public users
Casetext [14]	AI-assisted semantic legal search engine	Improves legal research through context based search and citation mapping	Operates under U.S. law; subscription cost restricts use in low income regions
CourtListener [15]	Open-access database of court decisions	Promotes legal transparency through free access to judicial records	Offers raw data only; lacks AI driven search, summaries, or workflow tools

3.4 Conclusion

An overview of the current developmental legal systems shows that, despite the proven application of technology into the legal processes by the global platforms, including ROSS Intelligence, DoNotPay, and Clio, they still serve the foreign jurisdictions and professional firms of the developed world. Though such systems are indicative of the simplification that AI and automation may bring to research, documentation, and client management, it is impractical in terms of high costs, regional incompatibility, and context irrelevance of these systems in Pakistan. IntelliLaw overcomes these shortcomings in providing an AI-supported legal assistance and case management platform designed with the Pakistani legal framework in mind. IntelliLaw bridges the gap between advanced yet inaccessible technologies and makes it locally accessible, simplifying legal interactions, informed decision making, and increasing transparency in the dispensation of justice.

Chapter 4 Software Requirement Specifications

This section will provide in detail all associated software requirements of our project. It includes the list of features, the functional requirements, the quality attributes, the non-functional re-quirements, related assumption, the hardware and software requirements, the use cases, graphical user interface, all necessary diagrams and figures, along with the risk analysis of our project IntelliLaw.

4.1 List of Features

The major features of the IntelliLaw are listed in this section. They are listed below:

- AI-based Legal Assistant Chatbot to aid lawyers and educate the general public about laws.
- Centralized Dashboard displaying case statistics, hearing schedules, and key metrics
- Comprehensive Case Management with create, edit, filter, and calendar scheduling options
- Invoicing System for tracking of payments, revenue, and generation of invoices
- Application-specific case fields and structured form for capturing comprehensive case information
- Authentication for secure access: role-based for attorneys, paralegals, and admins

4.2 Functional Requirements

Besides the basic registration and login, IntelliLaw, as a web application, provides the following features:

- The system shall allow users to create, view, edit, and delete cases.
- The system shall display a dashboard with real-time case and hearing statistics.
- The system shall provide a search, filter, and sort functionality for case records.
- The system shall offer calendar views (day, week, month) for hearing and case scheduling.
- The system shall allow users to upload and manage case attachments.
- The system shall enable custom field creation for case-specific data.
- The system shall support client and opponent contact management.
- The system shall generate and manage invoices with tax, discount, and total calculations.
- The system shall store and display payment history and remaining balances for cases.
- The chatbot shall provide legal assistance and information retrieval for lawyers and laypersons.

- The system shall implement secure login and role-based user access control.
- The system shall maintain responsive design for desktop and mobile users.

4.3 Quality Attributes

The quality attributes form an important aspect of the performance evaluation of the software system. They provide a check-list to help in judging quality of the software. The quality attributes of our system are:

4.3.1 Usability

The application offers a friendly, clean interface that lets attorneys and laypersons use the navigation intuitively, generate cases, gain insights from dashboards, and have the capacity to interact with chatbots without technical knowledge.

4.3.2 Reliability

The system allows consistency in performance as well as data integrity, reducing incidences of down-time and avoiding the infiltration of case records or details of the invoices, also during peak usage.

4.3.3 Scalability

It is made in order to support growing workloads and allows law firms to manage an increasing number of users, cases, and financial records without performance degradation.

4.3.4 Security

A strong authentication, role-based access control and encryption mechanisms are in place to protect sensitive information, such as client information, case files, and payment

4.3.5 Availability

The web system is developed in a way to guarantee high uptime and reliability by ensuring that the users have access to their data and operations from anywhere at any time.

4.3.6 Performance

The application guarantees fast response time on all the important aspects of the user operation, including case search, loading of the dashboard, as well as chatbot responses, for a smooth user experience

4.4 Non-Functional Requirements

Our system non-functional requirements are discussed here

- Most of the common functionalities for instance, the user authentication, case forms and invoice generation should be designed to be reusable components which will make the code run efficient in Skadi while ensuring consistency all the way.
- Operational smoothness and rapid reaction times shall be ensured: all important actions, e.g. case search, loading of the dashboard, response of the chatbot, shall be equal to 2 seconds. It should be able to support up to 500 concurrent users with no noticeable degradation in performance
- The system shall be easy to extend, with new modules, such as a court hearing tracker, document automation, or reporting tools, that can be slotted in without major refactoring.
- The system should conform to the standard data protection and privacy laws regarding the managements of legal cases, ensuring client confidentiality and data integrity is maintained
- Ensure encryption of all types of user data, including case files and financial records, with SSL/TLS; ensure role-based access control and appropriate authentication. Passwords and sensitive data should be stored securely after hashing.

4.5 Assumptions

It is assumed that all users will have a stable internet connection and use modern web browsers compatible with the application; this also applies to attorneys, paralegals, and firm staff. The system assumes a secure server environment that has enough storage and processing capacity to efficiently manage simultaneous users and data transactions. Users of the system are expected to enter accurate information while creating cases, invoices, or client records. Besides that, basic computer literacy is assumed for all users, and the organization should provide the necessary credentials and permissions for authorized personnel only. Apart from that, the AI-powered chatbot shall assume access to updated legal databases or APIs to create relevant and accurate responses.

4.6 Use Cases

Use cases included in this section are those from the use-case model that represent some significant, central functionality of the final system, or that have a large architectural coverage—they exercise many architectural elements—or that stress or illustrate a specific, delicate point of the architecture.

Table 4.1: Use case for authorizing users

Name	User Login		
Actors	Attorney, Paralegal, Admin		
Summary	Enables authorized users to securely log in to the application		
Pre-Conditions	User must have valid credentials.		
Post-Conditions	User gains access to the dashboard.		
Special Requirements	Passwords must be hashed and validated securely.		
Basic Flow			
Actor Action		**System Response**	
1	User enters username and password	2	System validates credentials.
2	User clicks Login.	4	System authenticates and redirects user to dashboard.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Incorrect email or password entered.

Table 4.2: Use case for creating new case

Name	Create New Case		
Actors	Attorney, Paralegal		
Summary	Allows users to create a new case with details, client information, and attachments.		
Pre-Conditions	User must be logged in		
Post-Conditions	New case record is stored in the system.		
Special Requirements	All mandatory fields must be validated before saving.		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Create Case button.	2	System displays a multi-section case creation form.
2	User submits the form.	4	System generates unique Case ID and saves record.
Alternative Flow			
3	User skips mandatory fields.	4-A	System highlights missing inputs and prevents submission.

Table 4.3: Use case for viewing, searching, filtering, and managing existing case records.

Name	View and Manage Cases		
Actors	Attorney, Paralegal, Admin		
Summary	Enables users to view, edit, filter, or print case records.		
Pre-Conditions	Cases exist in the database.		
Post-Conditions	Updated or filtered cases are displayed.		
Special Requirements	Real-time filtering without page reload.		
Basic Flow			
Actor Action		**System Response**	
1	User navigates to Cases tab.	2	System shows case list in table format.
2	User clicks Edit Case.	4	System opens editable form for selected case.
Alternative Flow			
3	User applies invalid or empty filters.	4-A	System displays “No Records Found.”

Table 4.4: Use case for viewing upcoming hearings in monthly, weekly, or daily calendar formats

Name	View Calendar and Hearing Schedule		
Actors	Attorney, Paralegal		
Summary	Provides calendar, week, or day views to track upcoming hearings.		
Pre-Conditions	Cases must have hearing dates.		
Post-Conditions	Hearing schedule is displayed.		
Special Requirements	Calendar updates dynamically.		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Calendar View.	2	System displays monthly calendar with hearings.
2	User switches to Week or Day View.	4	System shows corresponding schedule.
Alternative Flow			
3	No hearings scheduled for period.	4-A	System shows “No Hearings Scheduled.”

Table 4.5: Use case for new invoice creation and storage for a client or case

Name	Generate Invoice		
Actors	Attorney, Paralegal		
Summary	Creates a new invoice linked to a client and optionally a case.		
Pre-Conditions	At least one client and case record must exist.		
Post-Conditions	Invoice is stored and visible in the invoice list.		
Special Requirements	Invoice number must be unique.		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Create Invoice.	2	System displays invoice form.
2	User selects client, enters items, and submits.	4	System calculates totals and saves invoice
Alternative Flow			
3	User omits required fields.	4-A	System prompts completion of missing details.

Table 4.6: Use case for viewing, filtering, and printing invoice records

Name	View and Manage Invoices		
Actors	Attorney, Admin		
Summary	Allows users to view, filter, and print invoices.		
Pre-Conditions	Invoices exist in the system.		
Post-Conditions	Invoices are displayed or printed.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User navigates to Invoices tab.	2	System lists all invoices.
2	User filters by paid/unpaid.	4	System updates table accordingly.
Alternative Flow			
3	No invoices found for filter.	4-A	System shows “No Records Available.”

Table 4.7: Use case for AI-powered chatbot interaction for legal advice and awareness

Name	Use Legal Assistant Chatbot		
Actors	Attorney, Layperson		
Summary	Allows users to interact with the chatbot for legal queries or awareness.		
Pre-Conditions	Chatbot service is active.		
Post-Conditions	User receives appropriate legal information.		
Special Requirements	Requires internet access and AI model availability.		
Basic Flow			
Actor Action		**System Response**	
1	User opens chatbot window.	2	System loads chat interface.
2	User types a legal question.	4	Chatbot processes query and responds.
Alternative Flow			
3	Chatbot cannot answer query.	4-A	System suggests related laws or human consultation.

Table 4.8: Use case for managing contact information for clients or opponents

Name	Manage Contacts		
Actors	Attorney, Paralegal		
Summary	Enables users to add or edit client, opponent, or lawyer contacts.		
Pre-Conditions	User must have access to contact management.		
Post-Conditions	Contact details are updated or saved.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Add Contact.	2	System displays contact form.
2	User enters name, phone, email.	4	System validates and saves contact.
Alternative Flow			
3	User leaves a required field blank	4-A	System highlights missing field.

Table 4.9: Use case for managing payment records and fee-related information

Name	Payment History Update		
Actors	Attorney, Accountant		
Summary	Updates payment records for a case or client.		
Pre-Conditions	Case and invoice must exist.		
Post-Conditions	Payment history is updated.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User opens Payment History dialog.	2	System lists past transactions.
2	User adds new payment entry.	4	System saves and recalculates outstanding amount.
Alternative Flow			
3	User enters invalid payment amount.	4-A	System displays validation error.

Table 4.10: Use case for managing case archiving and restoration

Name	Archive or Restore Case		
Actors	Attorney, Admin		
Summary	Allows archiving of old cases or restoring them when needed.		
Pre-Conditions	Case exists and user has permission.		
Post-Conditions	Case status changes to Archived or Active.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Archive Case.	2	System confirms and updates case status.
2	User views archived list.	4	System shows all archived cases.
Alternative Flow			
3	User selects Restore Case.	4-A	System reactivates case and moves it to active list.

Table 4.11: Use case for user logout and session termination

Name	Logout		
Actors	Attorney, Paralegal, Admin		
Summary	Safely logs the user out of the system.		
Pre-Conditions	User must be logged in		
Post-Conditions	User session is terminated		
Special Requirements	All session data should be cleared securely.		
Basic Flow			
Actor Action		**System Response**	
1	User clicks Logout.	2	System ends session and redirects to login page.
Alternative Flow			
3	Network failure during logout.	4-A	System displays “Logout Failed. Please Retry.”

4.7 Hardware and Software Requirements

List the hardware and software requirements that will be required to develop and deploy the project.

4.7.1 Hardware Requirements

The computer should have at least an Intel Core i5 or equivalent processor for seamless development and execution of the Next.js application. At least 8 GB RAM with 256 GB SSD storage could ensure smooth running of IDEs, databases, and local servers. A stable internet connection is a must for installing dependencies, accessing APIs, and deploying the project on cloud platforms.

4.7.2 Software Requirements

The following will get you started with this project: a modern operating system like Windows 10, Ubuntu 20.04+, or macOS that supports Node.js and Git. Next.js is used for building the application, Node.js serves as the runtime, and MongoDB or PostgreSQL is used as the database. Other important tools include Visual Studio Code for coding, Git for version control, and Postman for API testing. You can deploy on either Vercel or AWS.

4.8 Graphical User Interface

GUI sample of the system is as follows:

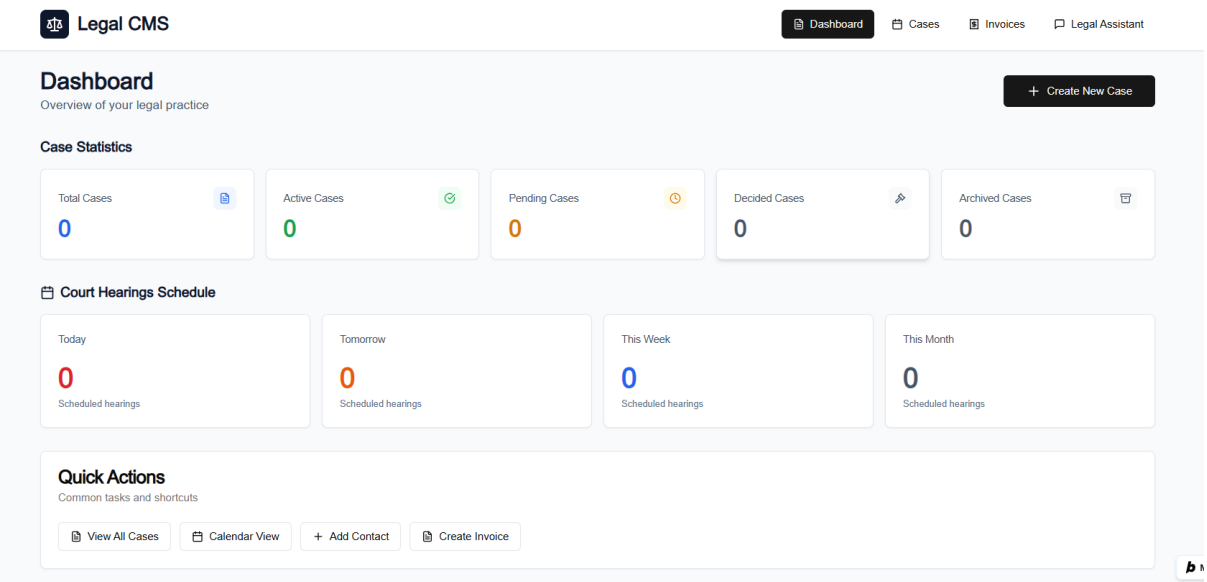

Figure 4.1: DashBoard

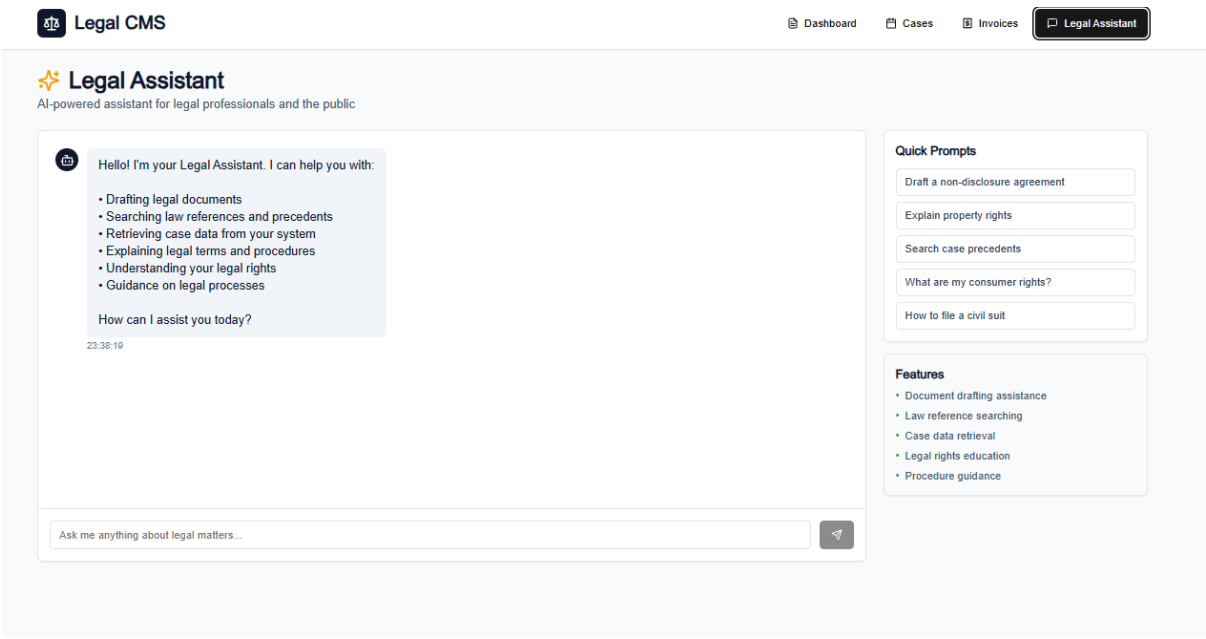

Figure 4.2: Legal Assistant

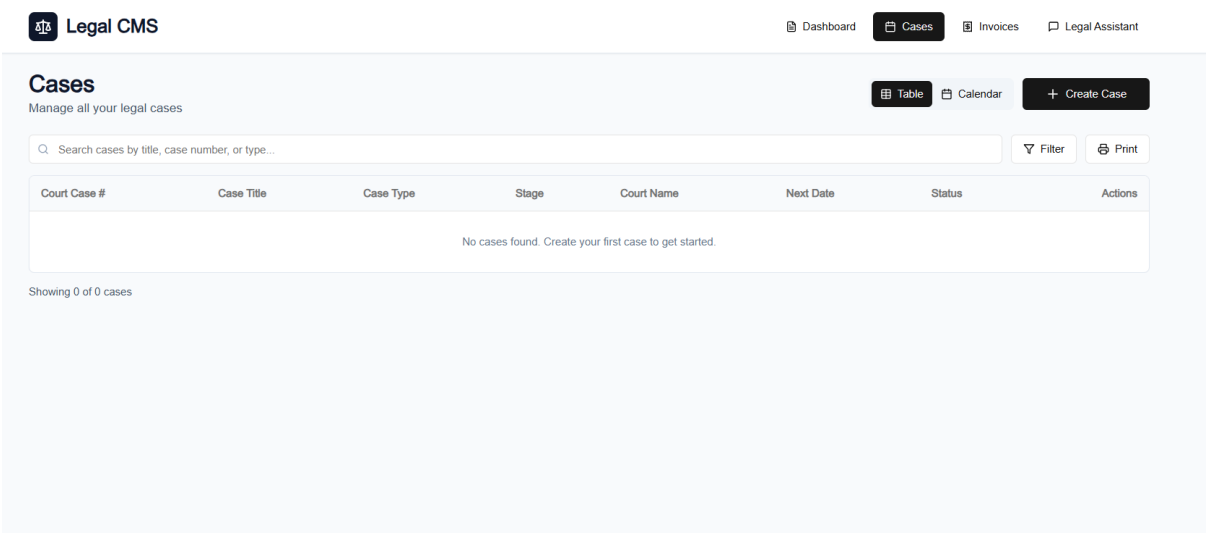

Figure 4.3: Create Case

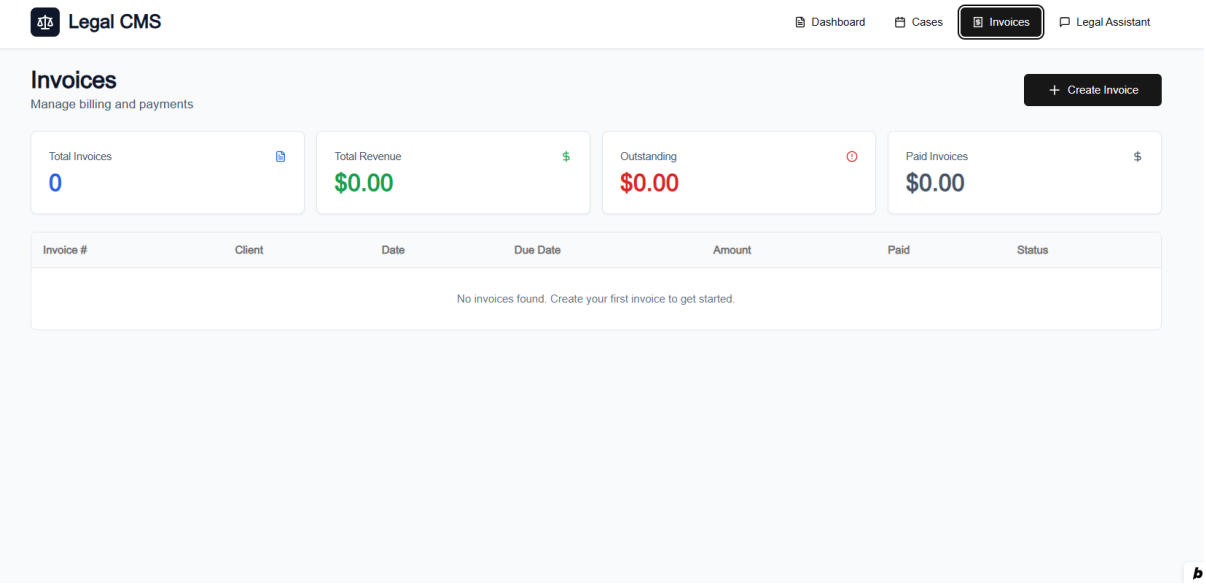

Figure 4.4: Invoice Management

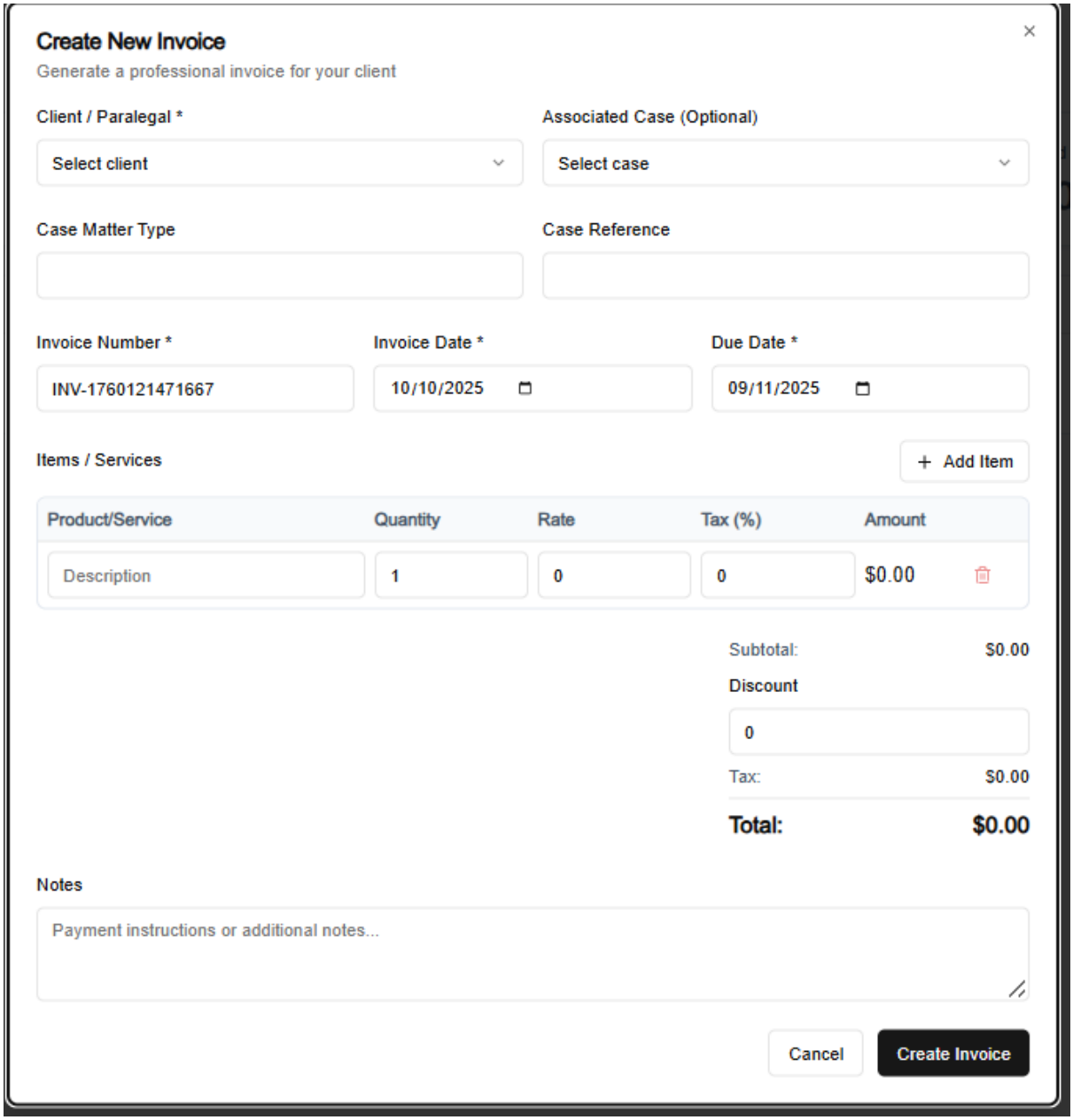

Figure 4.5: Invoice Form/Edit

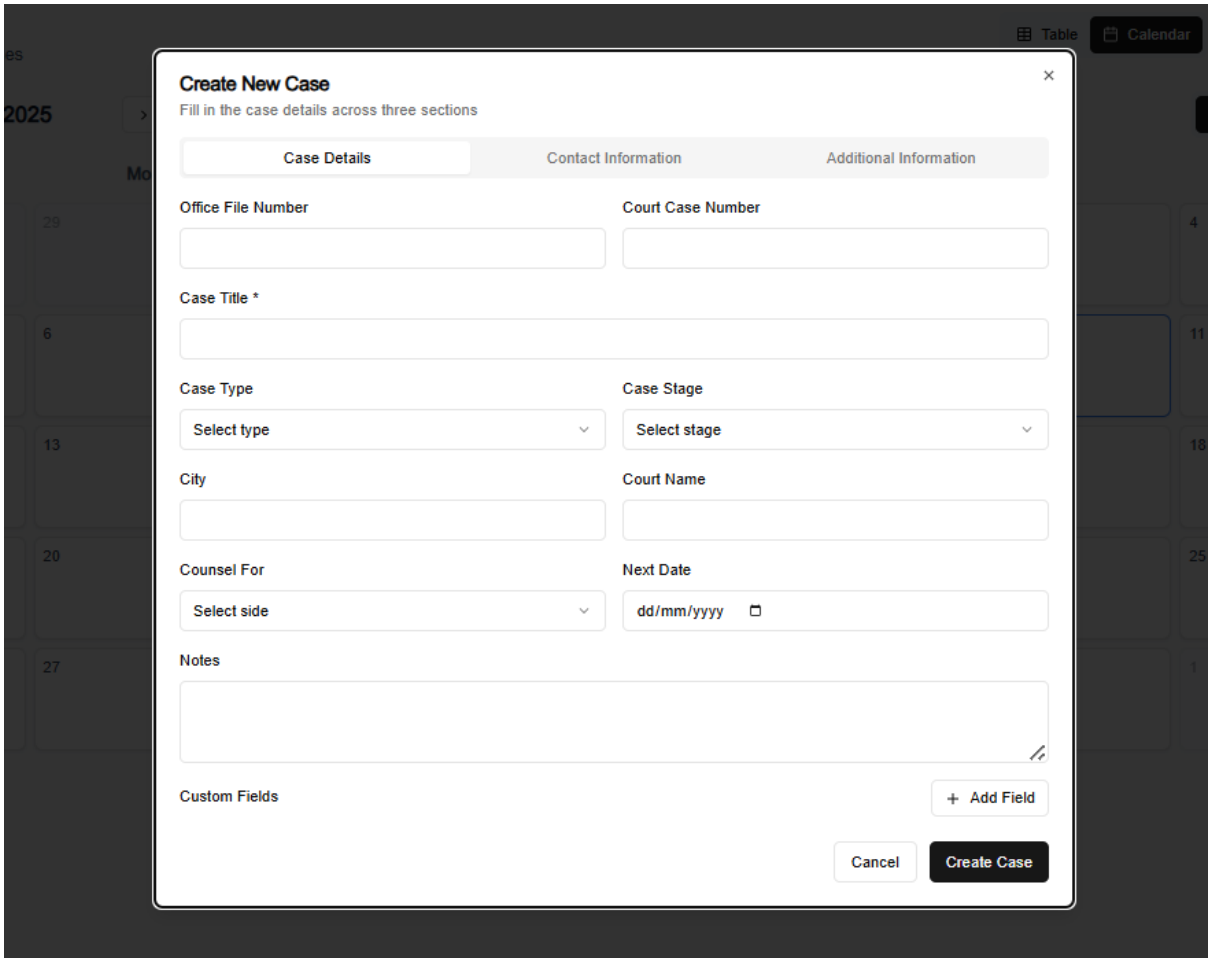

Figure 4.6: Case Form

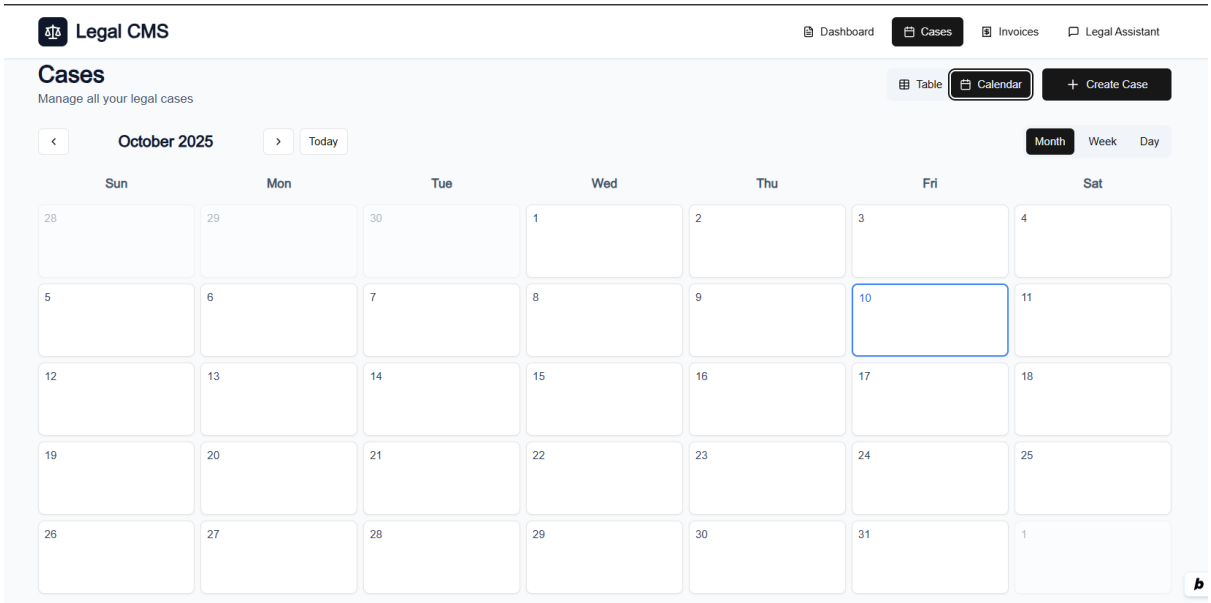

Figure 4.7: Calender View

4.9 Database Design

4.9.1 ER Diagram

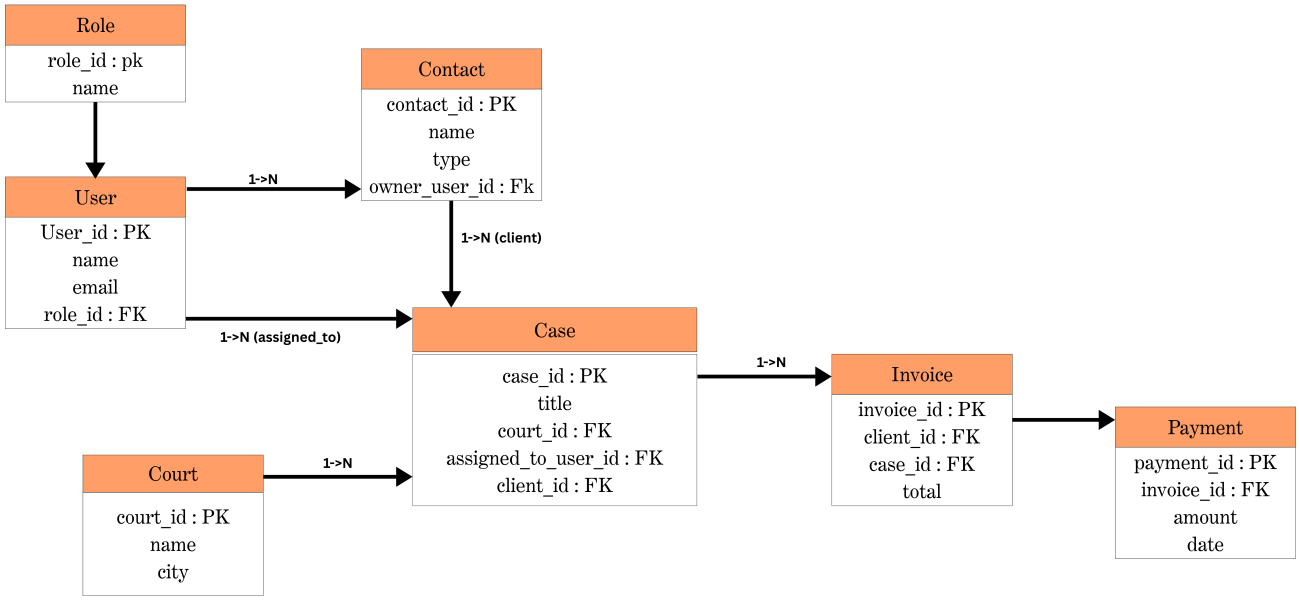

Figure 4.8: ER Diagram

4.9.2 Data Dictionary

This section provides the data dictionary for the Case Management System for Attorneys and Law Firms. Each table describes the entities, their attributes, data types, descriptions, and constraints.

Table 4.12: Data Dictionary - Admin

Entity	Attribute	Data Type	Nullable	Description
Admin	admin_id	Int	No	Unique ID
	name	String	No	Admin name
	email	String	No	Email
	password	String	No	Password
	phone	String	No	Phone
	created_at	Datetime	No	Created time

Table 4.13: Data Dictionary - Attorney

Entity	Attribute	Data Type	Nullable	Description
Attorney	attorney_id	Int	No	Unique ID
	name	String	No	Attorney name
	email	String	No	Email
	password	String	No	Password
	specialization	String	No	Expertise area
	phone	String	No	Phone
	created_at	Datetime	No	Created time

Table 4.14: Data Dictionary - Client

Entity	Attribute	Data Type	Nullable	Description
Client	client_id	Int	No	Unique ID
	name	String	No	Client name
	email	String	No	Email
	phone	String	No	Phone
	address	String	Yes	Address
	created_at	Datetime	No	Created time

Table 4.15: Data Dictionary - Case

Entity	Attribute	Data Type	Nullable	Description
Case	case_id	Int	No	Unique ID
	title	String	No	Case title
	description	Text	Yes	Case detail
	status	String	No	Case status
	client_id	Int	No	Client ref
	attorney_id	Int	No	Attorney ref
	created_at	Datetime	No	Created time

Table 4.16: Data Dictionary - Document

Entity	Attribute	Data Type	Nullable	Description
Document	doc_id	Int	No	Unique ID
	case_id	Int	No	Case ref
	file_name	String	No	File name
	file_path	String	No	File path
	uploaded_at	Datetime	No	Upload time

Table 4.17: Data Dictionary - Appointment

Entity	Attribute	Data Type	Nullable	Description
Appointment	appointment_id	Int	No	Unique ID
	case_id	Int	No	Case ref
	client_id	Int	No	Client ref
	attorney_id	Int	No	Attorney ref
	date	Datetime	No	Meeting date
	status	String	No	Status

Table 4.18: Data Dictionary - Payment

Entity	Attribute	Data Type	Nullable	Description
Payment	payment_id	Int	No	Unique ID
	case_id	Int	No	Case ref
	amount	Float	No	Amount
	method	String	No	Pay mode
	date	Datetime	No	Pay date
	status	String	No	Pay status

Table 4.19: Data Dictionary - Message

Entity	Attribute	Data Type	Nullable	Description
Message	message_id	Int	No	Unique ID
	sender_id	Int	No	Sender ref
	receiver_id	Int	No	Receiver ref
	content	Text	No	Message body
	sent_at	Datetime	No	Sent time

4.10 Risk Analysis

Some of the technical and business risks that could be associated with system development are described below

4.10.1 Technical Risks

The system may encounter compatibility issues between frontend Next.js and backend API or database connection failures. There is also a potential for performance related issues in case of increased concurrent users. May lead to system down-time, data loss, or slow system performance.

4.10.2 Business Risk

The final system may not meet user requirements, or there is over-spend of budget/time due to feature creep. May lead to project delays or rejection by stakeholders.

4.11 Conclusion

This SRS document describes in great detail the functional and non-functional requirements of the system. This therefore forms a comprehensive reference for developers, testers, and stakeholders who will be involved in this project to ensure that there is a mutual understanding of the scope and objectives of this project before the development phase commences.

Chapter 5 High-Level and Low-Level Design

This chapter presents the overall design of the IntelliLaw system, including both high-level architecture and detailed component-level design. It also explains how different modules interact, what technologies are involved, and the logical flow of data inside the system. The objective is to offer clarity on how IntelliLaw's conceptual model translates into a practical, working, and implemented system.

5.1 System Overview

IntelliLaw is the AI-equipped legal assistance and legal case management system that can upgrade the access to justice, automate repetitive jobs performed by legal professionals and provide reliable legal information to citizens. The system comprises of two major modules, an AI-based Legal Chatbot and a Case Management System, both integrated in a single web system. At a high level, IntelliLaw is based on a modular client server architecture that integrates Natural Language Processing (NLP) techniques, structured databases, and a Web user interface The back-end AI One solution is an API that ensures a process that is in the same order and format as a long standing user process, yet one that is unfair as the queries are not processed in time.One solution is an API that ensures an intuitive and acessible user interface by modelizing the users query and generating documents, while the front end ensures the security of the user interface since the query is not processed in time. sible experience for al the generals population and the legal practitioners.

5.1.1 Purpose and Functionality

The primary purpose of the IntelliLaw is two fold:

1. It is intended to give preliminary legal advice to the general public by responding to common legal questions and referencing relevant sections of Pakistani laws, assisting the users to make informed decisions.
2. For professionals in the legal field, it automates case tracking, document management, and administrative workflows (allows lawyers and firms to improve productivity and keep their digital records.

5.1.2 System Organization

The architecture for IntelliLaw is three-tier architecture, which separates the system into three logical tiers:

1. Presentation Layer (Front-end):

- Provides the graphical interface for the users that include, lawyers, firms, and citizens.
- Allows the interaction via chat, form submissions, and dashboards.
- Handles data visualization such as case updates, invoices and reminders.

2. Application Layer (Backend):
   - Implements business logic and system flow work.
   - Manages the processing of the chatbot's NLP model, user requests and case data.
   - Integrates API for real-time updates of laws and manages the communication between the frontend and database.
3. Data Layer (Database):
   - Stores user information, case information, etc. Document and conversation logs.
   - Ensures that data is consistent, confidential and that it can be quickly retrieved. both structured SQL and unstructured NoSQL storage based on the requirements

5.1.3 Basic Design Approach

Modularity, scalability and security are some of the principles used in the design of the system. Case management Both voice assistant and chatbot modules are independent but communicate through a common backend. This modularity allows for

- Easier updates, such as retraining the chatbot without affecting the case module.
- Improved and updated fault tolerance.
- Simplified scalability for handling large datasets and multiple users.

Given the highly sensitive nature of the legal data, security and privacy have been very deeply entrenched in the design. Through the system protects user's confidentiality by role-bases authentication, data encryption, and access management following ethical AI regulations.

5.1.4 Overview of Data Flow

One of the typical interaction flows of IntelliLaw system is as follows:

1. The user can be either a citizen or a lawyer who interacts with the frontend web interface to the system.
2. The request that can be a legal query or simply case entry, for example request is sent via secure RESTful APIs to the backend..

3. For chatbot queries, the backend places the text input into the NLP engine, which pulls up relevant information received from the legal dataset and generate an output.
4. In case of management operations, the backend interacts with the database for the and of storing, updating or getting case details.
5. The result of the processing is sent back to the frontend and presented to the user.

IntelliLaw combines artificial intelligence with new web technologies in order to simplify and improve how people interact with the legal system, and By pairing an AI-powered chatbot and a case management platform, it helps citizens access reliable legal information easily while lawyers are able to manage their work more effectively. Its modular and layered design offers a secure and scalable and adaptable framework which means that the people and the legal professionals can place confidence in IntelliLaw accurate, timely and well-organized legal assistance.

5.2 Design Considerations

The important factors, which affect the design and execution of the IntelliLaw system, are described here. These include the design goals and design concepts used as a guide, as well as the approaches used throughout the development,

5.2.1 Assumptions and Dependencies

During the design phase of IntelliLaw, there were several assumptions and dependencies identified to ensure consistent and realistic implementation and These are:

- It is assumed that IntelliLaw will be deployed on a modern web-based infrastructure for standard web browsers such as Chrome, Edge and Firefox. The backend depends on Python-based frameworks, such as Flask or Django, for API management and communication with the AI model
- The system assumes stable internet connectivity for end users and hosting servers, which have enough computational resources to support NLP model inference and data storage.
- It is assumed that users both citizen and legal professionals have basic computer literacy and be able to interact through web application interface for submitting queries, handling cases and uploading documents.
- The chatbot is dependent on access to data sources about law and APIs, for example, Pakistan Code and Supreme Court archives, to bring latest legal information and references.
- The functionality of the design can be updated periodically regarding the social datasets and NLP

models. The system architecture supports modular upgrades with no major re-engineering.

5.2.2 General Constraints

Whilst designing IntelliLaw, we faced a number of technical and practical limitations that directly influenced the way the system was built and the manner in which it performs. Some of these were from the environment in which the system runs and some others were associated with security, performance and available resources.

5.2.2.1 Hardware and Software Environment

The system is reliant on a stable internet connection And also, a server that has enough RAM, CPU to run the model of the chatbot and the cases. Limited hardware resources may cause model responses and file uploading to be slow, particularly if there are many users are active.

5.2.2.2 End-User Enviroment

IntelliLaw is mainly browser-based as such, one can expect that performance and layout may vary a bit for different devices and/or browsers. Users on slow or older browsers might experience some small delays to access dashboards or chats.

5.2.2.3 Resource Availability

Since we are dealing with limited student-level infrastructure the depends on common hosting or universities provided with machines. Heavy training or long-term data storage is not sensible in this environment, and so lightweight deployment is prioritized in the design.

5.2.2.4 Interoperability and Integration

The system is integrated with external APIs for legal data. These APIs could change in format or become unavailable, disrupting the process of automatic updates. To reduce this risk, we used flexible JSON-based handling so small API changes will not crash the system.

5.2.2.5 Security and Privacy

Legal information should be handled with care. The system enforces login authentication and encrypted passwords. controlled access to data. Any security breach could result in loss of trust; thus, security is a constraint at the design level, as opposed to an afterthought.

5.2.2.6 Performance Limitations

The chatbot and database should be able to give a quick response preferably within a few seconds. Longer or more complex queries however may take more time when the AI model processes legal text. Caching and optimizing queries were included in order to reduce delays.

5.2.2.7 Network Communication

Everything is communicated through https. If the network fails or the server is unreachable, the chatbot and dashboard will not be able to work. Offline functionalities were taken into consideration out of scope for this version.

5.2.2.8 Data Storage and Distribution

The database is used to store profiles for users, the cases, and the uploaded documents. Storage capacity is limited so only important files and compressed records are kept. Backup procedures are planned but not automated in the first version.

5.2.2.9 Testing and Validation

Due to limited testing resources, only functional and unit tests were performed. Full-scale load or penetration testing was not possible in the current set up, but the design provides for the addition of those later

These constraints were boundaries that moved us in the right direction to those things that truly matter. They shaped IntelliLaw into a system that's practical, efficient and maintainable even if working with limited

5.2.3 Goals and Guidelines

While designing IntelliLaw, we followed a few main goals and simple design rules that helped us stay focused. These goals were not only technical, but also about keeping the project manageable and user-friendly. The following points highlight the key ideas that guided our design choices.

1. **Simplicity:** Our team tried to follow the "Keep It Simple" rule throughout development. The aim was to make the system intuitive, easy to use, and easy to maintain. Instead of adding too many complicated features, we built a few important ones that actually work well such as the chatbot for legal queries and the case management dashboard. Keeping things simple also made it easier for new users to get comfortable with the system without needing tutorials.

2. **Modularity and Reusability:** We broke down the system into distinct modules, such as the Chatbot Module, Case Management Module, Authentication Module and Database Layer Each part han- dles its own tasks and makes it easier to update or fix through APIs one module without any effect on the rest of the system. This way, future teams are also able to reuse portions of our code in other projects, if necessary.

3. **Security and Data Privacy:** Since IntelliLaw is dealing with legal data and client data, one of For example, ”our number one priorities was to ensure everything was secure”. We included login authentication, password encryption, and restricted access to users based on their roles Security not been treated as an add-on; it was built into the design from the beginning, both in the backend and database structure.

4. **User-Centered Design:** We wanted IntelliLaw to be useful not only to lawyers but even to regular citizens who may not have a legal knowledge. For that reason, the interface is clean and straightforward. Buttons, forms and chat prompts are simple, and the chatbot's responses are written in a simple language that the users can understand them easily.

5. **Scalability and Performance:** Another design goal was to ensure that the system could handle more users and more data in the future. We did not hard code the limits and used a modular backend so that things like analytics or document verification could be added on later We also focused on response time, ensuring that queries and case operations are done effectively even when there are multiple users are active.

6. **Maintainability and Team Collaboration:** As IntelliLaw was a team project, we ensured the design enabled all the people to work on different parts at the same time; Using GitHub for version control helped us to effectively track changes. We also wrote some internal documentation for our modules to enable future students to easily understand and extend

7. **Realism and Feasibility:** Finally, we made our goals realistic. We only included features that could be done within the time frame and the hardware available in the semester. This helped us deliver a stable system rather than leaving incompleteness as the components. Each design decision was made with time and practicality in mind.

Together, these principles allowed us to design IntelliLaw as a reliable, easy-to-use platform that can change over time because new features and requirements are added

5.2.4 Development Methods

The approach followed for the development of IntelliLaw was flexible and iterative- as in the Agile methodology. This approach wasfit for our small team size and for us to manage the project in short, manageable stages rather than doing everything at once Each stage was targeted on a different part of the system, such as the chatbot, the case management dashboard or the database connection. We started off with simple prototypes, testing them internally, then refining the prototypes by including feedback from our supervisor/team discussions. This step by step process enabled us to identify problems early and make small improvements instead of waiting until the end to fix major issues. We had incremental development, in which after finishing one module, we integrated and tested it with the others. For example, when the chatbot interface was up and running we linked it to the backend APIs and database before moving on to case Management features. This ensured that every component worked well with the rest of the system; Version control was managed through GitHub, so that all team members were able to work on their assigned Modules without overwriting each other's codes. Regular commits and pull requests kept the progress going organized and traceable. This also made it easy to roll back changes if something broke. The AI and the backend design process was designed according to research-based development. We studied similar systems, trained and fine-tuned a language model with a legal text, and developed APIs for queries and responses. The web interface has been developed with the database schema to ensure the smooth flow of data between components. For testing, unit testing and functional testing were done at each stage. Each new feature was checked by independent individuals, then after integration; Keeping the feedback loop short helped us quickly identify and correct bugs/design problems. In the end this method of development ensured we were organized and made it easier to release small, working versions regularly. It also helped us to be flexible and deal with any issues as they arose instead of following a rigid plan.

5.3 System Architecture

The architecture of IntelliLaw is based around simple independent modules that work together to deliver a fully legal support and case management platform. Each part of the system does its specific role and communicates with other components using secure APIs. This configuration makes the system easier to maintain, update and scale as new features or improvements are made

At a fundamental level, IntelliLaw is built in three major subsystems with a small number of common services that glue everything together:

1. **Chatbot Service:**Processes legal queries and handles user interactions.

2. **Case Management Service:** Assists lawyers in handling cases of clients and case management documents.
3. **Model Serving Service:** Hosts and runs AI models responsible for language understanding and legal content generation.

Supporting components like **API Gateway**, **Authentication Service** and **Database Layer** con- connect these subsystems in order to maintain data consistency and guarantee security across the platform

Major Responsibilities

- **User Interaction:** Enables user to chat, ask questions related to law, upload or view case-related documents.
- **AI-Driven Support:** The chatbot understands the queries made by the user and replies with the correct responses using The AI models that are hosted on the Model Serving Service.
- **Case Management:** Allows lawyers to see client cases, create invoices and schedule court dates.
- **Security & Access Control:** Every request is passed through a secure authentication layer to ensure authorized access.
- **Data Handling:** ecures case data, chat logs, and legal documents in database and file storage.

Component Overview

- **Frontend:** Provides the main interface for the public users as well as legal professionals. It includes the chat interface, case dashboard, and uploading documents.
- **API Gateway:** Acts as the central point to which user requests are sent, validating and routing them to the appropriate service.
- **Chatbot Service:** Accepts user queries and generates AI-based replies and fetches relevant legal information.
- **Case Management Service:** Manages client data, case data, documents, billing & reminders
- **Model Serving Service:** Deploys the AI/ML models and inference APIs for chatbot operation-tions.
- **Authentication & Authorization:** Deals with login, registration and role-based access control.
- **Database & Storage:**Stores the structured data like Cases, Users and Chat logs and unstructured data that have legal documents and attachments.

System Workflow

When a user sends a message or uploads a document the frontend sends the request via the API Gateway as per directs it to the proper service such as Chatbot for queries or Case Manager for streamlining case, etc. Data operations service (ment) If required AI processing, the Chatbot Service communicates with the Model Serving Service Generating Response The Database Layer is used to store and retrieve information as needed.

This workflow helps ensure that all interactions are logged, validated and interactively processed free of overload- ing any single component.

Rationale for Design Choice

At the beginning, monolithic structure was considered for simplicity reasons. However, it was replaced with a modular architecture because of the following reasons:

- Each module is free to update or scale individually.
- The AI model can be run on different GPU servers without any impact on other parts.
- It permits concurrency of development and testing of the project by different developers.
- Easier to maintain, debug and expand as time goes on.

This architecture is based on the microservices architecture and promotes flexibility, scalability, and clear separation of concerns. Common patterns used are:

- **MVC:** for frontend structure.
- **Repository Pattern:** for clean database access.
- **API Gateway Pattern:** for routing and security.
- **Adapter Pattern:** for integrating external APIs.

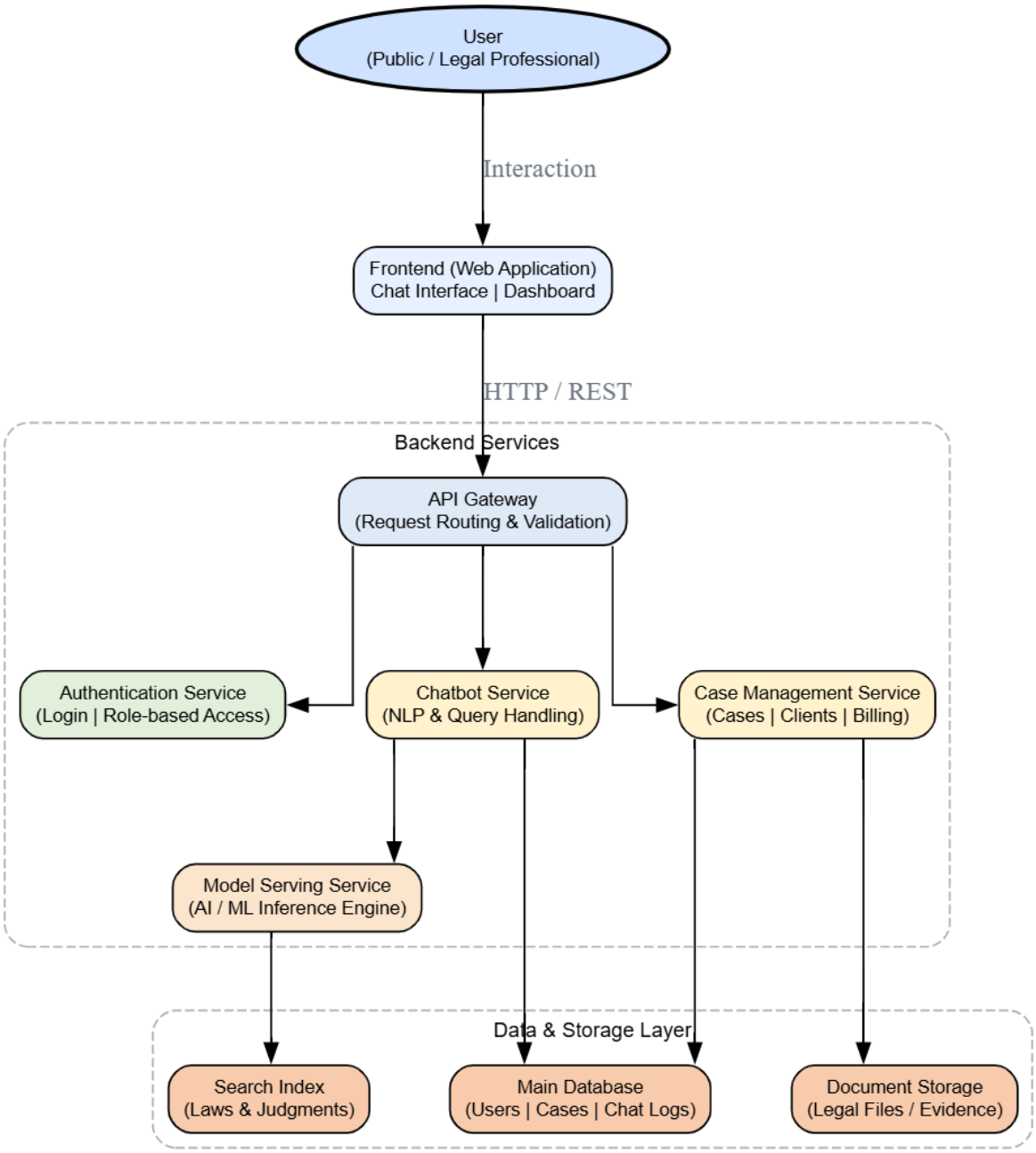

Figure 5.1: High-level System Architecture of IntelliLaw.

5.3.1 Subsystem Architecture

This section uses internal structure description of IntelliLaw's main subsystems and how their internal components work together.

A. Chatbot Service

The Chatbot Service is responsible for user conversations and creating legal responses.

- **Input Handler**: Accepts messages from the user and also checks for validity before processing the messages.

- **NLU (Natural Language Understanding)**: Detects the intent and entity extraction from the text.
- **Knowledge Retriever**: Retrieves related laws/ judgements from the legal DB.
- **Response Generator**: Generates human readable responses and drafts documents using AI models
- **Session Manager**: Keeps track of the context of the conversation over a series of interactions.
- **Audit Logger**: Only logs interactions for QA and retraining.

B. Case Management Service

This subsystem is intended to process the workflow of lawyers in a smooth manner.

- **Case Controller**:Manages the data creation, update and retrieval of cases using the services provided by the REST APIs.
- **Document Handler**: Handles uploading, file storage and retrieval of case-related documents.
- **Billing & Invoice Generator**: Automates fees and creates payment receipts.
- **Scheduler**: Manages things such as court reminders, deadlines, email notifications etc.
- **Access Control Module**: Makes sure that only lawyers are able to access their allotted cases.

C. Model Serving Service

The Model Serving Service hosts and manages AI and ML models that are used for natural language processing known as LLMs.

- **Inference Engine**:sed to run AI models (such as transformer or NER models) to interpret the legal queries.
- **Preprocessing Pipeline**: Cleans and preprocesses text for model inference
- **Postprocessing Module**: Refines model output to produce comprehensible, response-aware of context.
- **Model Manager**: Deals with loading, versioning and updating of AI models.
- **Cache Layer**: The layer that stores the frequent responses to enhance the speed by preventing the redundant computation.

D. Data and Storage Layer

This layer is used to store all persistent and reference data.

- **Relational Database**: Stores user profiles, case details and logs
- **Object Storage**: Stores large documents Such as Affidavits or evidence of case

- **Search Index**: Supports full-text search for the legal documents and the case files.
- **Cache**: One way of improving response time is to temporarily store common queries.

Overview of Subsystem Interaction

1. The user input is sent to the API Gateway from the Frontend.
2. The Gateway routes queries about chat to the Chatbot Service or requests related to the case to the Case Man- agement Service.
3. The Chatbot communicates with the Model Serving Service to produce Artifical Intelligence responses.
4. The services both use Database Layer for storage.
5. Accreditation is centrally controlled by the Auth Service.

5.4 Architectural Strategies

The architectural design of IntelliLaw is informed by a mixture of practical development goals and long period of system requirements. Major strategies are on the basis of modularity, scalability, maintainability, and security while ensuring that each component will be able to evolve independently while still contributing to the general workability of the system. Each and every design decision was made after taking different performance, complexity and development effort are in fact trade-offs. The selected strategies are also matched with the goals described above: keep the system simple, extensible and reliable.

5.4.1 Strategy 1 – Modular Service-Based Architecture

One of the important strategies adopted in the case of IntelliLaw is a service-based modular architecture. Instead of building of one monolithic application, the system is broken down into smaller independent modules such as the Chatbot Service, Case Management Service, and Model Serving Service.

Each module has a fixed set of responsibility and offers his or her services to others through the RESTful APIs.

Reasoning:

This strategy was utilised as it increases the maintainability and scalability of the application. When one module needs updates, e.g., when it is to upgrade the model of the AI then, it can be redeployed and modified without affecting other services. It also enables the members of the team to work on different modules at the same time and So, it increases development speed and collaboration.

Trade-offs:

- **Pros:** Separation of concerns, Parallel development, Easier debugging, Independent scaling.
- **Cons:** More setup is needed for inter-service communication and deployment coordination

Alternatives Considered:
A development of a monolithic design was considered at first but this is easier to develop and deploy. However, since it would have massive impact on limited flexibility and future updates would be cumbersome especially the ones involving AI model upgrades were rejected.

Related Patterns:
It is in line with the design patterns that we have now, such as Microservices, API Gateway, and Repository Patterns, that help in keeping a clean interface between different parts of the system.

5.4.2 Strategy 2 – Layered Design and Technology Choices

Another important strategy was to structure IntelliLaw into these three major layers:

- Presentation Layer
- Application Layer
- Data Layer

It will ensure logging clarity and data flow

Layer Descriptions:

- **Presentation Layer:** The web-based front-end is the part of the system where the user interacts with the system. It concentrates on the response and user experience.
- **Application Layer:** Deals with business logic, routing requests, and communication between mod- ules. It comprises of services like Chatbot, Case Management and Model Serving.
- **Data Layer:** It is responsible for the safe storage and retrieval of structured and unstructured data, such as user profile, case records and legal documents.

Reasoning:
This layered architecture maintains the architecture neat and easy to maintain. Each layer can evolve independently. For example the frontend interface can be redrawn without modifying logic on the backend.

- **Frontend:** React.js for responsive and modular UI
- **Backend:** Node.js with Express (for lightweight, scalable APIs)

- **Database:** MongoDB (flexible document-based storage)
- **AI Model:** Transformer-based model integrated through Flask or FastAPI
- **Authentication:** JWT-based authentication provides secure access

Trade-offs:

- **Pros:** High modularity, code reusability, and easier testing.
- **Cons:** Introduces minor overhead in the interlayer communication.

Alternatives Considered:
Also, using a one layer architecture, meaning direct interaction of the UI with the database, was considered and rejected because the presentation and data logic would be tightly coupled making any future maintenance hard.

5.4.3 Strategy 3 – Secure and Fault-Tolerant Communication

Given the sensitivity of legal data, IntelliLaw is built with security and fault tolerance in mind for intern and external communications.

Key Mechanisms:

- JWT based authentication for validating each and every API Request. data between client and server encrypted with HTTPS/TLS
- Centralised logging and error handling that will hopefully begin to identify and resolve issues early.
- Automated recovery mechanisms retry logic, for example, to keep uptime in the event of temporary failures.

Reasoning:
These measures help to make confidential legal information secure and make the system reliable even when the usage is high or/and when partial failures occur.

Trade-offs:

- **Pros:** Ensured high data integrity and trustworthiness
- **Cons:** Slightly increased computational overhead as a result of encrypting and validating.

Alternatives Considered:
Simpler token based authentication without encryption was avoided, as this would not meet the necessary security standards for how data will be handled to serve legal interests.

5.4.4 Strategy 4 – Future Extensibility and Reusability

The architecture is designed to support long-term future extensions without any major rewrites, so IntelliLaw can prolonged for a good period of time.

Implementation:

- Modular components allow adding new features, such as voice based query support or court case analytics.
- The layer gets to be isolated allowing for replacement by newer and better-AI models.
- Common services such as Authentication, Database Access, and Notification services can be shared by other modules in the future.

Reasoning:
This approach allows continuous improvement without interfering in the operation of existing users or workflows and will help also in achieving the objective of maintainable and scalable platform.

5.4.5 Concluding Remarks on Architectural Strategies

IntelliLaw achieves a balance between the architectural strategies of flexibility, maintainability and perfor- mance. By combining modular service-based design, layered design, and robust mechanisms for security, the system is flexible enough for future use and meets current functional requirements These choices ensure that IntelliLaw is robust enough for real world legal use and scalable enough to grow with time.

5.5 Domain Model/Class Diagram

The IntelliLaw domain model is used to define the major entities and relationships in the system. Provides a structural overview of how information flows, and how different components interrelate at the data level; The model represents users, legal professionals, cases, documents, chat interaction, and AI generated responses, all of whom work jointly to offer legal assistance and case management functions.

5.5.1 Domain Overview

The system has two main types of entities:

1. **User-related Entities:** These will represent all the individuals who interact with the system, like citi- zens and legal experts.
2. **Case and Interaction Entities:** This includes everything related on case, handling of documents,

chat conversations, and AI powered recommendations.

The domain model is specifically designed to allow for scalability and maintainability. Each class handles specific responsibilities and engages others through well-defined relationships, for example one-to-many (e.g. one lawyer managing many cases) or many-to-many (e.g. multiple users with multiple chat sessions).

5.5.2 Description of Major Classes:

Here is class description table

Class Name	Description
User	Represents both citizens and legal professionals. Stores basic details, credentials, and user roles.
Lawyer	A specialized type of user with additional fields such as license number, specialization, and active cases.
Citizen	Represents a regular user who can ask legal questions or submit case requests.
Case	Contains details of legal cases, including case ID, description, client information, status, and hearing schedule.
Document	Stores metadata for uploaded files such as contracts, petitions, and court documents. Each document belongs to a case.
ChatSession	Represents a conversation between the user and the AI chatbot. Contains timestamps, queries, and responses.
Message	Stores individual messages within a chat session (text, attachments, or AI replies).
AIResponse	Holds responses generated by the AI model for a specific query. Linked to the ChatSession.
Invoice	Generated for billable legal services. Contains billing amount, issue date, and payment status.
Notification	Handles alerts and reminders such as hearing updates or payment due notifications.
Admin	Oversees the system, manages user accounts, and monitors data activity.

Table 5.1: Description of Major Classes

5.5.3 Overview of Relationships

- A User can have several ChatSessions.
- A Lawyer can be involved in numerous Cases, and each Case can contain a number of Documents and Invoices.
- Every Case is one belonging to one Citizen (the client) and one Lawyer (the representative).
- Every ChatSession might have a number of Messages and some messages can have an AIResponse attached to them.
- Cases Notifications occur for both Lawyers and Citizens depending on the events in the case.

5.5.4 Class Diagram

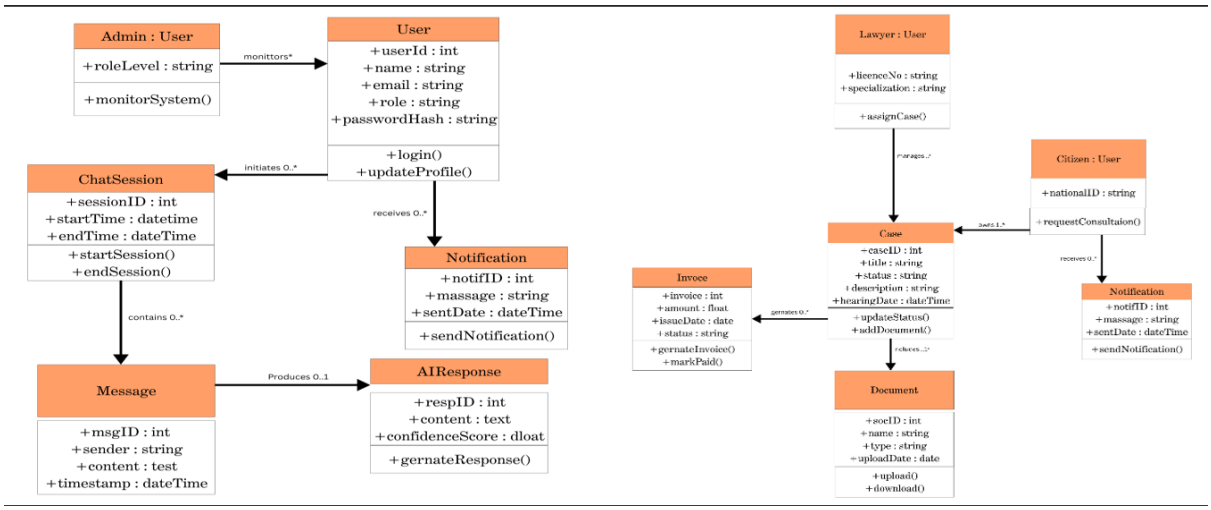

Figure 5.2: Class Diagram

5.6 Policies and Tactics

This section describes design policies and implementation tactics used during the development of IntelliLaw. Such decisions might not have strong influence on the overall architecture of the system but are highly influential in the way the modules are implemented, tested and updated; The aim of such policies is to Since there are many reasons why code should be simplified, optimized, and simplified, ensure the consistency in coding-style, reliability in performance, and simplicity in updates in the future.

5.6.1 Policy 1: Coding Standards and Naming Conventions

Description:

A consistent coding standard was also followed for the development tasks in the entire project. All the backend modules were written using Node.js Python with clean code principles and meaningful naming conventions - followed in both teams

Implementation:

- Variable and function names should be camel cased in JavaScript and snake cased in Python.
- Class names follow PascalCase (e.g. UserProfile, CaseManager).
- Comments in code which explain the logic in simple English.
- Each module has a README file for quick understanding of the functionality of the modules.

Reasoning:

Consistent coding style makes code more readable and easier to work with as a team. It also makes onboarding new developers easier.

Alternatives Considered:

We considered using the standard configuration for linter, something like ESLint + Prettier auto formatting, but chose to do the formatting by hand because the project is quite small and managed by a limited team.

5.6.2 Policy 2: Error Handling and Logging

Description:

Proper error-detection and logging mechanism were put in place to easy debug and ensure that system failures did not impact the experience of the users..

Implementation:

- All API endpoints have try-catch blocks.

- Every service has its error codes and messages.
- Errors are being logged using a centralized logger (like Winston) in Node.js.
- Exceptions that are expected but not anticipated are used to send notifications to developers in testing environments.

Reasoning:

The advantage of this approach is errors are isolated, traceable and recoverable without crashing the system thus it improves the maintainability and reliability of the application.

Alternatives Considered:

We looked at using an external monitoring tool such as Sentry, but we didn't want to do that because it would only be able to monitor all environments during test time. poslite tuvieron for academic deploments.

5.6.3 Policy 3: Version Control and Collaboration

Description:

All code was maintained using Git and GitHub which made collaborative development and proper version tracking. Features, bugs and documentation were committed to separate branches to preserve the integrity of the project.

Implementation:

- **Main Branch:** Always kept stable and deployed.
- **Feature Branches:** sed for development of specific modules, such as feature / chatbot or feature / case- api.
- **Pull Requests (PRs):**Used for code reviews and integration.
- **Commit Messages:** Followed the format type(scope): message (e.g., fix(auth): token expiration issue).

Reasoning:

With version control everything is traceable, the collaboration is painless and it's easy to roll back when a feature causes any problem.

Alternatives Considered:

We've taken into consideration local backup based on files, but eventually opted for using GitHub for much better collabora-, tion, remote stowage and potential of CI/CD integration..

5.6.4 Policy 4: Testing and Validation Approach

Description:

Reliability and correctness is ensured in this project by unit testing and functional testing.

Implementation:

- **Unit Tests:**Written with pytest jest, for python node js respectively.
- **Integration Tests:** hese are conducted for validating the back-end modules communication with the database.
- **Manual Testing:**UI features are working on different browsers to check for their compatibility.

Reasoning:

Testing first snapped bugs that's what caught them before integration. This approach helps to ensure that new changes don't break existing functionalities.

Alternatives Considered:

Fully automated CI/CD testing pipelines were considered but not implemented because of limited time and resources.

5.6.5 Policy 5: Directory Structure and Code Organization

Description:

The source code has an organized directory structure to keep it clean and separate the concerns

Example Structure:

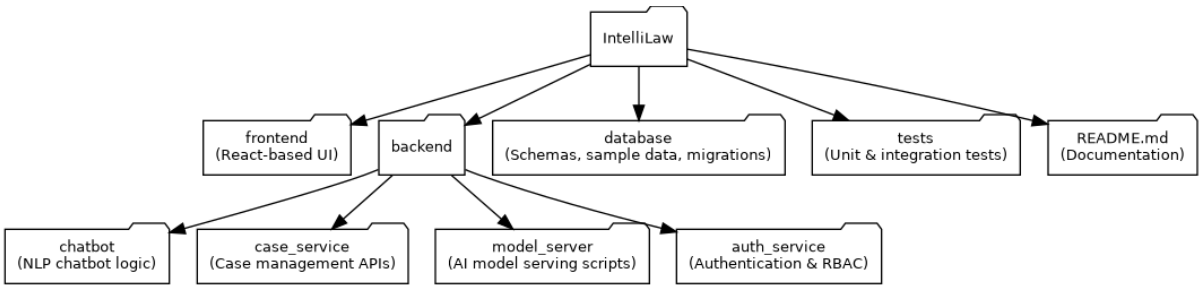

Figure 5.3: Directory Structure

Reasoning:

This organization makes it easier to find files, debug modules and deploy services independent.

5.6.6 Policy 6: Build and Deployment Strategy

Description:

Instead of complicated CI/CD pipelines, simple build scripts for deployment and testing have been employed. The For the backend services, Docker containerization was used which ensures a stable environment during the development and testing.

Implementation:

- Dockerfile for each of the microservices.
- Environment variables in the form of .env files manage secret information.
- Local testing Using Docker Compose

Reasoning:

Containerization makes the system portable and mitigates those ”it works on my machine” problems. It Also aids in scalability in case the system needs to be moved on a cloud platform later.

Alternatives Considered:

Directly deploying and deploying on a shared hosting server was considered here but dropped due to environment inconsistas. tencies.

5.6.7 SUM TOTAL

The above design policies and tactics will render the IntelliLaw to remain structured, reliable, and easy to maintain. Consistent coding practices, version control practices, testing practices, and deployment practices as- sure the stability of the system in the long run including the short run. These practices do not alter the archi tecture of it but very much enhance the general quality and sustainability of the development of the system.

5.7 Conclusion

In all, this chapter expanded on the high and low level designs of the IntelliLaw system: how its components work together to offer intelligent legal assistance and case management func- tionalities. Such a modular and layered architecture provides the benefit of scalability, maintainability and ease of enhancements in the future. Collectively, these design elements establish a good foundation for the implementation and integration of system in the next phases.

Chapter 6 Implementation and Test Cases

This chapter basically captures how the main parts of IntelliLaw was implemented when we (then) got beyond the design phase. Much of the real work occurs here, because of many things that appear simple on diagrams behave very differently when you start wiring them together in code.

Instead of frontal or obvious routes like in the past chapters, this chapter talks about the internal logic, background algorithms, and the hidden procedures keeping the system running in the background For instance, how the AI pipeline determines what documents to retrieve, how the retrieval engine prioritizes legal sections user access validation backend does how to store records securely.

Along with that this chapter also explains how we tested these components in a realistic way to ensure assures that each and every feature works both in normal and edge case scenarios. The goal is to provide a better insight on the actual mechanisms that cause the working of the system rather than just describing it's visible features.

6.1 Implementation

The IntelliLaw implementation phase is where the system started getting its shape beyond dia- grams and conceptual models. Since our project is a combination of a traditional web platform combined with an A.I.- assisted legal engine ,we had to divide the implementation work into different layers and handle each component that has a slightly different approach. Most of the development was carried out by means of a modular struc- ture so that each service, model interfacing, case management, authentication, and data handling retrieval could be created independently and then hooked up using well-defined APIs.

On the backend side, we took a rest based architecture, where every module has a series of endpoints that communicate with the database or other service internal to the system. The backend is implemented in Node.js and Express because it provides good flexibility and it will make integrating middleware like JWT based authentica-. tion, input validation, logging, and rate limiting. One of the most important aspects of the implementation was the rationale for the legal retrieval and response generation. For this we built a retrieval pipeline that takes in user queries, generates the semantic embedding and then matches it up against our pre-processed law corpus using vector search mechanism. We also maintain a fall keyword-based font also. back using BM25 style lexical search so that if the embedding match is ambiguous the system can still search related sections of law based on exact wording This hybrid retrieval design guarantees that the system doesn't place too much reliance on one technique and can provide stable results even when there is a variation in the user input.

The integration of the AI component is via an API layer where the backend coordinates the communi-

cation between the retrieval module and the language model.Once the relevant sections in the laws are identified, we create a structured prompt, which consists of the retrieved passages, metadata, and system instructions. This prompt is then passed to the model through the provider's API (for example, OpenAI or a com - patible inference server). The output of the model is cleaned, issues regarding format are checked, and then passed back to the frontend. In order to prevent hallucinations, we had added a rule based filter: in case the gener- ated answer does not actually refer to the retrieved documents, the system will re dé query or to safe fallback message. For the case management and user-related features we've applied a structure based on a relational database using PostgreSQL. Normalizing tables where we have data such as clients, cases, documents, invoices, re- minders can be linked neat using foreign keys was done. We even encrypted sensitive fields and added role-based access logic at API and database level in order to prevent unauthorized access. On the frontend, we used React to implement an interface, which would update dynamically, both according to user roles and responses from the system. The chatbot interface, of course, needed some additional wiring, especially in streaming reply and clean citation show. So far, implementation efforts have been concentrated on the construction of the basic foundation of the system: the retrieval-augmented AI assistant, secure user and cases handling modules and the basic workflow of document uploads and the processing of legal enquiries. These components are the backbone on which all other features will build and provide sufficient stability to move on into more complex features during the next phase.

6.1.1 Implementation of Retrieval Component

The retrieval module is the backbone of IntelliLaw where the quality of what the AI answers is highly contingent on what documents it pulls out before generation. In implementation, we made attempt to design this module in a manner that would combine semantic comprehension and traditional keyword relevance. This hybrid setup helped us deal with legal data, usually very sensitive to exact wording, but requiring deeper contextual matching. Some of the key algorithms and steps applied inside the retrieval pipeline are provided here.

1. **Embedding Generation**

   In order to form a system that knows what people are asking, in depth-difference than just words- we first convert the text into numerical embeddings.
   The process goes as follows: this:

   (a) A pre-trained sentence embedding model (similar to the architecture used in SBERT) is used.

   (b) Each section from the law, each paragraph, fragment of judgment, is pre-processed (clean- ing, normaliza- tion) before embedding.

(c) Once the embeddings are created during the preparation of the dataset, we store them and don't re- processing at runtime.

(d) At query time, the user's text is converted in the same way to an embedding vector.

Numeric representation to measure the similarity between the queries and passage in law. more "semantic" way, and not just key words.

2. **Vector Database** After generating embeddings, we stored them inside a lightweight vector search system.

   (a) We implemented a (flat or HNSW) FAISS-like index (depending on available memory)

   (b) Each entry stored the following:

      - vector
      - document ID
      - section reference
      - metadata: (court, year, title)

   (c) The index enables fast nearest neighbour search, a prerequisite if the size of the dataset inches of millions. creases.

   (d) We have prepared different indices for the statutes and judgments so that retrieval can be tuned with respect to query type.

   The aim was to permit fast top k queries even in the presence of multiple parallel request.

3. **BM25 Fallback**

   Semantic search alone cannot cover all the aspects of legal text, therefore we created a lexical fallback system.

   (a) We built a BM25 index out of chunks of text that we have cleaned helps when:

      - The user enters specific section numbers
      - Legal keywords appear
      - Embeddings don't work because phrasing is rare Both BM25 and vector results are considered separately

      the BM25 and the vector answers are considered separately.

   (b) One method yields poor results and other compensates

In practice this fall-back helped a lot for Pakistani law queries where there is a lot of variation between English/Urdu phrasing.

4. **Search Ranking (Hybrid Scoring)**

   Once both the results of the vector and BM25 modules are available, we rank them using a combined scoring mechanism.

   (a) Vector similarity score is equal to cosine similarity of embeddings

   (b) BM25 score is equal to lexical relevance

   (c) Final ranking is equal to weighted combination (we tuned these weights during testing)

   (d) The top-k results after ranking are selected and packaged for the model.

   This hybrid approach made the system more stable and irrelevant retrieval cases are reduced.

5. **Query Processing Pipeline**

   The entire retrieval pipeline works in the order of:

   (a) **Receive Query**

   - The raw user query is cleaned (lowercase, stopword removal, punctuation handling).

   (b) **Detect Query Type**

   - The raw user's query gets cleaned (lower case, stopword removal, handling of punctuations).

   (c) **Generate Embedding**

   - Query is passed through the embedding model.

   (d) **Vector Search**

   - Retrieve top-k matches from FAISS index.

   (e) **BM25 Search (Parallel)**

   - Retrieve lexical matches from the BM25 index.

   (f) **Combine & Rank**

   - Merge results, resolve duplicates, apply weighted ranking.

   (g) **Return Top Context**

   - Selected text chunks in addition with metadata are passed to the prompt builder for the AI model.

This structured pipeline make sure that the model always which had a direct impact on the accuracy of answers.

6.1.2 Implementation of AI Response Generator

AI response generator is basically the part which is responsible for converting the legal text, which is retrieved, to meaningful answer. Our focus in this regard was on keeping it simple, while maintaining reliability, Because legal information cannot be handled so loosely. Here is the list of the important pieces we implemented.

1. **Prompt Engineering**
   (a) We cluster retrieval top passages into an organizational prompt.
   (b) The prompt contains the following: the user query, sections of laws that are relevant, and a brief instruction that asks the model to remain factual.
   (c) We also add some little format guide so that the answer will be readable and this includes cite- tions naturally.
2. **API Integration**
   (a) The constructed prompt will be sent to the AI model using the backend via the progression of a Rest API.
   (b) We keep the request down by sending top-k passages only.
   (c) A timeout and retry logic was introduced since sometimes the model delays or drops the request.
3. **Post-Processing Logic**
   (a) Extra spaces or unnecessary explanations are cleansed from the model output.
   (b) We then extract the final text along with any text referenced to section numbers.
   (c) If the answer appears incomplete and/or off-topic then the system can invoke a second retrieval and retry.
4. **Safety Filters**
   (a) A simple rule-based layer that will check whether the model outwardly provides legal advice accidentally.
   (b) Regarding anything risky that seems to come up in the question, we paraphrase the answer in a neutral way.

(c) A necessary disclaimer is included to the safety.

5. **Citation Enforcement**

   (a) We scan the model outputs to make sure that all the citations are the same as the retrieved passages.

   (b) Missing the citations in case citations, the system inserts the ones from ranked results.

   (c) The model is further requested with stronger requests, after wrong or hallucinated citations are blocked.

6.1.3 Implementation of Case Management Module

The case management module is the portion of the system that is responsible for processing all of the lawyer side workflows, such as client and case and document management. So we tried to keep it simple but organized so everything remains easily traceable and updating Following are the major things that we implemented in this module.

1. **User Roles**

   (a) We added role based access so that different people only see the ones that they are supposed to.

   (b) The Backend first checks the JWT token and role of the user before performing any action related to the case.

   (c) This is to avoid accidentally accessing another attorney's cases and/or documents.

2. **Case CRUD**

   (a) Lawyer can Create new cases, change details, or close case when its done.

   (b) Each case has associated with it one client and one attorney, using foreign keys.

   (c) Added some basic validation (no duplicate case number and required fields checking),

3. **Handling Documents**

   (a) Files that are uploaded within the case are much safer, stored in a secure folder or object storage.

   (b) Only the owner and the admin of the case can open / download the documents

   (c) For each file there is a record stored which is just a small metadata containing type, timestamp, and uploader.

4. **DB Schema**
   (a) The important tables that are present in this module are: Users, Clients, Cases, Documents and CaseNotes.
   (b) Relation kept simple- one lawyer has many cases, one case has many documents
   (c) Sensitive fields such as client contact information are stored using simple encryption.
5. **Error Handling**
   (a) We also added checks for missing fields, invalid IDs or unauthorized access to cases.
   (b) In case of failure of an operation, a friendly message is returned instead of raw server error.
   (c) Upload failures and corrupted files are caught up in time from causing issues in database.

6.1.4 Implementation of Authentication & Security

The authentication and security part was something we didn't want to take lightly with IntelliLaw because system is dealing with legal data personal client information. Among the implementation, we tried to implement a simple but also robust configuration in terms of user identity, data protection and rules of access Below are the major pieces that we built into this module.

1. **JWT (JSON Web Tokens)**
   (a) System generates on login JWT that stores the id of the user and his/her role.
   (b) The token is returned to frontend and sent together with every subsequent request.
   (c) Backend takes care of verifying this token before allowing any protected action like opening a case or uploading a document.
   (d) Tokens also have an expiry time to mitigate the risk if someone tries to leaks it.
2. **Encryption**
   (a) Sensitive fields like client phone numbers or mails are stored as encrypted in the database.
   (b) We used basic AES style encryption so even if somebody does gain access to db, the raw values are still unreadable to them.
   (c) Passwords are NOT stored but they are directly hashed into a secure hashing function.
3. **Access Control**
   (a) Introducing role-based checks We have added role-based checks where different users only get to see their allowed features.

(b) Lawyers can only access cases they work on, admins get wider access.

(c) Unauthorized access attempts are met with a simple ”not allowed” message instead of disclosing any internal errors.

4. **API-Level Protection**

   (a) Each protected route has middleware that is able to validate the JWT.

   (b) The middleware also verifies whether the user’s role does match the required permissions.

   (c) Rate limiting has been added to login routes to prevent brute force attack:

   (d) Input validation filters added to prevent malformed and suspicious requests.

5. **Audit Logging**

   (a) Key actions include logging case edits, document uploads, and login attempts.

   (b) Logs store timestamp and user ID so issues can be traced if needed.

   (c) This helps us track unusual behavior or potential security problems later.

6.1.5 Frontend Interface Implementation

In the case of the frontend, our main issue was keeping the interface simple and clean so that users aren’t confused while working with the system. Most of the UI work but was done in React, because it is flexible and makes it easier to deal with changing states especially where the chatbot is producing responses or when forms are updated. We targeted especially the chat interface, the main dashboard, as well as different types of forms that lawyers use to create or update cases

1. **Chat UI**

   (a) The interface of the chat was made to feel natural just like a normal messaging application.

   (b) Messages are kept in state and rendered one by one such that the user can clearly see the conversation history.

   (c) While the AI is thinking, we showed a small ”typing. . . ” indicator to make the experience smoother.

2. **Dashboard**

   (a) oming hearings, plus the statistics for each of these across the state.(There is also a dashboard available to give a quick rundown of active cases and upcoming hearings, plus docu- ments.

   (b) We use simple cards and tables to make the intended data not look cluttered.

   (c) Dashboard automatically refreshed after some action like adding a case or uploading a file.

3. **Dynamic Forms**

   (a) To create and/or edit cases, we have used dynamic forms in which necessary fields are loaded based on user actions.

   (b) Form validation is done in real time so the mistakes can be corrected instantly.

   (c) The form changes automatically when there is a change in the type of case such as civil and criminal case fields.

4. **State Management**

   (a) We have used React hooks like useState, useEffect, for small interactions.

   (b) For bigger aspects, such as data for logged-in users or case details, we approved of context, to avoid prop drilling.

   (c) This keeps the code more neat as well as it avoids random UI bugs.

5. **Response Display Formatting**

   (a) AI responses are displayed with highlighted citations so that users are able to easily see where the information came from.

   (b) Long answers are divided into smaller and more readable blocks so as not to overwhelm the user.

   (c) Any error message or a fall-back response is styled a little bit differently, so it is easily noticeable

Chapter 7 Conclusions

IntelliLaw is an integrated AI-powered legal-assistance and case-management platform built to improve access to Pakistani laws for both the general public and legal professionals. The system combines a natural-language chatbot trained on Pakistani statutes and judgments with a secure web-based case management module that handles client tracking, document storage, invoices, and hearing reminders; thus, enabling citizens to get plain-language guidance while giving lawyers a lightweight productivity dashboard. This overall purpose and scope are described in the abstract and executive summary of the report.

On the technical side, IntelliLaw uses a modular, service-based backend with a hybrid retrieval pipeline: semantic embeddings plus vector search, FAISS with a BM25 lexical fallback, which are joined by a weighted ranking so that the model receives the most relevant legal passages. Those passages are structured into prompts and sent to an LLM for answer generation. The implementation focuses on prompt engineering, citation enforcement, safety filters to avoid giving direct legal advice, and a retry/post-processing loop to improve factuality. The retrieval and AI response pipeline and its design choices are detailed in the implementation chapter.

Role-based access control and JWT authentication, basic encryption for sensitive fields, API-level protections and audit logging, a React-based frontend with a conversational chat UI and dynamic forms, and Docker-based containerization for deployment. Testing so far focused on unit and functional tests with incremental/agile development; heavy load testing or penetration testing and automated backups were noted as out-of-scope for the first phase and flagged as areas for improvement.

Transition from MVP to a production-ready platform by increasing legal coverage, hardening reliability and security, and adding advanced features. Some concrete, recommended items for phase-2 are:

- Expand and maintain corpus: adding more current statutes, case law, and jurisdictions; introduce automated scheduled updates from official sources.
- Robust testing & ops: run load, stress, and penetration testing; add CI/CD pipelines, automated backups, and cloud deployment with autoscaling.
- Monitoring & analytics: build dashboards for real-time monitoring of uptime, query latency, retrieval quality, and user analytics to find failure modes and dataset gaps.
- Improve the quality and safety of models by adding human-in-the-loop evaluations, more sophisticated prompt/ensemble strategies, stricter hallucination, and fine-tuning on legal domains. New product features: multilingual support , document analysis (contract scanner & risk flags), lawyer directory/booking integration, case collaboration tools, and premium analytics for firms.

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